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FUNDAMENTAL ARITHMETICAL OPERATIONS

Percentage

A percentage is a mathematical concept that represents a number or ratio expressed as a fraction of 100. It is often denoted using the percent sign, "%".

Ex: 9% can be converted to a fraction as $9/100 = 0.09$

If, we want to calculate y% of x, then

$$y\% \text{ of } x = x \times \frac{y}{100}$$

Percentage Formula

Q. If 40% of P = 80, then find the value of P.

Sol: $P \times 40/100 = 80$

$$\Rightarrow P = 80 \times 100/40$$

$$\Rightarrow P = 200$$

Fraction and Percentage

To express x% as a fraction

$$\text{We have, } x\% = \frac{x}{100}$$

$$\text{Thus, } 20\% = \frac{20}{100} = \frac{1}{5}; 48\% = \frac{48}{100} = \frac{12}{25} \text{ etc.}$$

Q. 2.4 converted to percentage is

Sol: To convert a number into percentage it will be multiplied by 100.

$$= 2.4 \times 100 = 240\%$$

∴ Required percentage is 240%

Q. 9/50 converted to percentage is:

Sol: To convert it in percentage, multiply it by 100

$$\Rightarrow (9/50) \times 100 \Rightarrow 18\%$$

∴ Required percentage is 18%

Fraction to percentage conversions table

Percentage	Fractions
50%	$\frac{1}{2}$
$33\frac{1}{3}\%$	$\frac{1}{3}$
25%	$\frac{1}{4}$
20%	$\frac{1}{5}$
$16\frac{2}{3}\%$	$\frac{1}{6}$
$14\frac{2}{7}\%$	$\frac{1}{7}$
$12\frac{1}{2}\%$	$\frac{1}{8}$
$11\frac{1}{9}\%$	$\frac{1}{9}$
10%	$\frac{1}{10}$

Percentage	Fractions
$9\frac{1}{11}\%$	$\frac{1}{11}$
$8\frac{1}{3}\%$	$\frac{1}{12}$
5%	$\frac{1}{20}$
4%	$\frac{1}{25}$
$66\frac{2}{3}\%$	$\frac{2}{3}$
75%	$\frac{3}{4}$
$37\frac{1}{2}\%$	$\frac{3}{8}$
100%	1

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★ Important Points

Successive Increase / Decrease

Successive Increase

If some value is increasing at X% and then Y%, then combined increment can be given by

$$\left[X + Y + \frac{X \times Y}{100} \right] \%$$

Successive Decrease:

If some value is decreasing at X% and then Y%, then combined decrement can be given by,

$$\left[X + Y - \frac{X \times Y}{100} \right] \%$$

Q. Difference between two numbers is 38. If one number is decreased by 15% and another number is increased by 19% then the numbers obtained are equal. Find the sum of the two numbers.

Sol: Let the two numbers be A and B.

A is decreased by 15% and B is increased by 19%. According to statement,

$$A \times (85/100) = B \times (119/100)$$

$$\Rightarrow A \times 5 = B \times 7$$

$$\Rightarrow A/B = 7/5 \text{ Now,}$$

$$\text{Difference} = A - B = 7x - 5x = 2x = 38 \Rightarrow x = 19 \text{ Thus,}$$

$$\text{Sum} = A + B = 7x + 5x = 12x = 12 \times 19 = 228$$

Q. Two successive price increase of 10% and 10% of an article are equivalent to a single price increase of

Sol: Let the price of an article be 'x'.

$$\text{For 1st price increase of 10\% gives } x + 10x/100 = 1.1x$$

$$\text{For 2nd price increase of 10\% gives } 1.1x + (10 \times 1.1x)/100 = 1.21x$$

So, equivalent single price increase

$$\Rightarrow 1.21x - x = 0.21x$$

$$\therefore \% \text{ increase} = (0.21x/x) \times 100 = 21\%$$

💡 Shortcut Trick

Successive percentage increase = $\{a + a + (a \times a)/100\}$
Here, $a = 10\%$

$$\Rightarrow \{10 + 10 + (10 \times 10)/100\}$$

$$\Rightarrow \{20 + 1\} = 21\%$$

∴ Required increase in percentage = 21%

Q. 500 is first decreased by 20% and then it is decreased by 25%. What is the final value?

Sol: 500 decreased by 20%

$$\Rightarrow 500 - 0.2 \times 500 = 400$$

Now, 400 is decreased by 25%

$$\Rightarrow 400 - 0.25 \times 400 = 400 - 100 = 300$$

∴ Final Value is 300

📄 Alternate Method

First value = 500

First % decrease = 20% Second % decrease = 25%

$$\text{Net \% decrease} = (-20) + (-25) + (-20 \times -25)/100$$

$$\Rightarrow -40\% (\text{negative sign shows the decrease})$$

$$\therefore \text{Required value} = 500 \times 60/100 = 300$$

Comparison of Percentages

★ Important Points

If A is R% more than B, then B is less than A by $R/(100 + R) \times 100 \%$

Q. If A's income is 40% more than the income of B, then what percentage of B's income is less than income of A?

Sol: Let the income of B be 100

$$\Rightarrow \text{Income of A} = 140$$

B's income is less than the income of A by $(140 - 100)$

$$= 40 \text{ Required percentage} = \frac{40}{140} \times 100 = \frac{200}{7} = 28\frac{4}{7}\%$$

📄 Alternate Method

By using the formula $\left[\frac{R}{(100 + R)} \times 100 \right] \%$

$$\Rightarrow \left[\frac{40}{(100 + 40)} \times 100 \right] \% = 28\frac{4}{7}\%$$

★ Important Points

If A is R% less than B, then B is more than A by $R/(100 - R) \times 100 \%$

Q. If the income of A is 20% less than the income of B, then by how much per cent income of B is more than income of A?

Sol: Let the income of B be 100

The income of A will be 20% less than A i.e. 80 The income of B is more than A by 20

Percentage of income of B is more than A by $[(100 - 80)/80] \times 100 = 25\%$

∴ Percentage of income of B is more than A by 25%

Income and Expenditure

Monthly Income is basically the earning earned by a person for working throughout the month. Now if we multiply anyone's monthly income by 12, then we will get "Yearly income".

"Expenditure" means the total amount which a person spends from his total income for different purposes.

"Savings" means the remaining amount after spending an amount from Income.

★ Important Points

Income = Expenditure + Saving

Q. A's monthly income is Rs X, and his monthly expenditure is Rs. Y. Next year his income is increased by m% and expenditure increased by n%, then we have to find the % increase in his savings.

Sol: Saving = Rs (X - Y)

Income increase = Rs $(X \times m/100)$

Expenditure increase = Rs $(Y \times n/100)$

Saving increase = Rs $\{(X \times m/100) - (Y \times n/100)\}$

∴ % increase in savings = $[\{(X \times m/100) - (Y \times n/100)\} \times 100]/(X - Y)$

Q. The monthly income of a person was Rs 75000 and his monthly expenditure was Rs 40000 next year his income increased by 25% and his expenditure increased by 12%, the % increase in his saving was?

Sol: Income = Expenditure + Savings

$\Rightarrow 75000 = 40000 + \text{Saving}$

$\Rightarrow \text{Saving} = 35000$ Now,

Income increases by 25% then

Increase in Income = $75000 \times 25\% = 18750$, Expendi-

ture increased by 12% then

Increase in Expenditure = $40000 \times 12\% = 4800$

Increase in saving = $18750 - 4800 = 13950$

Percentage increase in savings

$\Rightarrow (13950/35000) \times 100$

$\Rightarrow 39.85\%$

∴ % increase in saving was 39.85%

⚡ Shortcut Trick

∴ % increase in savings = $[\{(X \times m/100) - (Y \times n/100)\} \times 100]/(X - Y)$

Here, X = 75000, Y = 40000, m = 25%, n = 12%

$\Rightarrow$ % increase in savings

$$= \frac{((75000 \times \frac{25}{100}) - (40000 \times \frac{12}{100})) \times 100}{35000}$$

= 39.85%

Numbers of students Present or absent in school

Q : In a class among x number of students, a% students are girls and the rest are boys. One day, m% of the total number of boys and n% of the total number of girls are present, then we have to find % of the total number of students who are absent.

Sol: Number of girls = $ax/100$

Number of boys = $x - ax/100 = (100x - ax)/100$

Total number of boys who are absent = $(100 - m)/100 \times (100x - ax)/100$

$\Rightarrow \{(100 - m)(100x - ax)\}/10000$

Total number of girls who are absent = $(100 - n)/100 \times ax/100$

$\Rightarrow ax(100 - n)/10000$

Total number of students who are absent = $\{(100 - m)(100x - ax)\}/10000 + ax(100 - n)/10000$

$\Rightarrow [x\{100(100 - m) + a(m - n)\}]/10000$

% of total number of students who are absent = $[x\{100(100 - m) + a(m - n)\}]/10000 \times 100/x$

= **$\{100(100 - m) + a(m - n)\}/100 \%$**



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Q. In a class 62.5% of the numbers of students are girls and the rest are boys if 80% of the number of boys and 50% of the number of girls are present, then what percentage of the total number of students in the class is absent?

Sol: Let the total number of students be 100

Percentage of Girls who are absent = $(100 - 50) = 50\%$
Girls who are absent = $62.5 \times (50/100) = 31.25$

Percentage of boys who are absent = $(100 - 80) = 20\%$
Boys who are absent = $37.5 \times (20/100) = 7.5$

Total number of boys and girls who are absent = $31.25 + 7.5 = 38.75$

Percentage of boys and girls who are absent = $(38.75/100) \times 100\% = 38.75\%$

$\therefore$ **38.75% students of the total class are absent.**

Shortcut Trick

% of total number of students who are absent = $\{100(100 - m) + a(m - n)\}/100 \%$

$$\frac{100(100 - 80) + 62.5(80 - 50)}{100}$$

= **38.75%**

Important Points

Marks based problems (Maximum or Passing marks)

Case 1:

Total mark of an exam paper is y. A student scored x% and failed by m marks.

Passing marks = $xy/100 + m$

Case 2:

Total mark of an exam paper is a. A student scored x% marks and failed by m marks. If she would score y% marks, she would obtain n marks more than the required number to pass.

$$ax/100 + m = ay/100 - n$$

$$\Rightarrow ay/100 - ax/100 = m + n$$

$$\Rightarrow a(y - x) = 100(m + n)$$

$$\Rightarrow a = 100(m + n)/(y - x)$$

$$\text{Passing marks} = ax/100 + m$$

$$\Rightarrow \text{Passing marks} = \{x/100 \times 100(m + n)/(y - x)\} + m$$

$$\Rightarrow \text{Passing marks} = \{x(m + n)/(y - x)\} + m$$

Q. Karan requires 40% marks to pass. If he gets 80 marks, he falls short by 20 marks. What was the maximum mark he could get?

Sol: $40\% \text{ of } x = 80 + 20$

$$\Rightarrow 40x/100 = 100$$

$$\Rightarrow x = (100 \times 100)/40$$

$$\Rightarrow x = 250$$

Maximum mark Karan could get is 250

Q. In an examination, Anita scored 31% marks and failed by 16 marks. Sunita scored 40% marks and obtained 56 marks more than those required to pass. Find the minimum marks required to pass.

Sol: Let maximum pass marks be x, then

$$\text{Passing marks} = x \times (31/100) + 16 \quad (1)$$

$$\text{Passing marks} = x \times (40/100) - 56 \quad (2)$$

From equation (1) and equation (2)

$$31x/100 + 16 = 40x/100 - 56$$

$$\Rightarrow 40x/100 - 31x/100 = 56 + 16$$

$$\Rightarrow 9x/100 = 72$$

$$\Rightarrow x = 72 \times (100/9)$$

$$\Rightarrow x = 800$$

$$\text{Passing marks} = 800 \times 31/100 + 16 = 248 + 16 = 264$$

Shortcut Trick

Let maximum marks $40\% - 31\% = (16 + 56)$

$$\Rightarrow 9\% = 72 \Rightarrow 1\% = 8 \Rightarrow 100\% = 800$$

$$\therefore \text{Passing marks} = 800 \times 31/100 + 16 = 264$$

Important Points

Venn Diagram based

What is a Venn diagram?

A Venn diagram is a picture that uses circles to show how different things or groups are related. Circles that cross each other show what they have in common, and circles that don't cross don't share those parts. Basically, Some problems are there which can be easily calculated with the use of a Venn diagram. We have described the method by taking an example which is shown below:



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Q: In an examination, 47% of students passed in science and 51% of students failed in mathematics. If 42% of students failed in both subjects, then what percentage of passed students in both subjects?

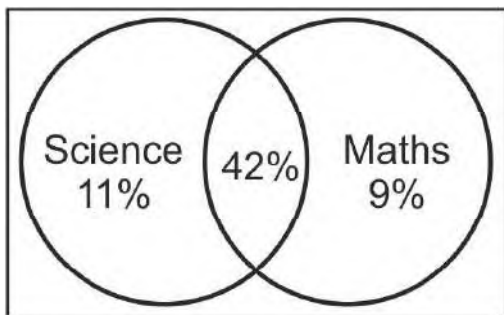
Sol:

Percentage of passed students in science = 47%
 Percentage of failed students in science = $100\% - 47\% = 53\%$
 Percentage of failed students in mathematics = 51%
 Percentage of failed students in science and mathematics both subjects = 42%

From the figure Total percentage of failed students = $11\% + 42\% + 9\% = 62\%$

∴ **Percentage of passed students**

= $100\% - 62\% = 38\%$



★ Important Points

Election or Vote based

Number of people in a village, among them $x\%$ didn't cast votes. The winner got $y\%$ of casted votes.

A number of people who didn't cast votes = $ax/100$

Number of people who cast votes = $a - ax/100$

Winner got $y\%$ of casted votes i.e. $(a - ax/100) \times y/100$

Q. In a village, 20% of people did not cast their votes. Ram won the election and he got 12.5% of casted votes which is 12,500 of the final total votes cast in the election.

Sol: Let the number of people be x
 20% who didn't cast their votes.

Number of people who didn't cast their votes = $20/100 \times x = x/5$

So number of people who casted votes = $x - x/5 = 4x/5$

Ram got 12.5% votes i.e. $(12.5/100 \times 4x/5)$ votes

⇒ Number of votes Ram got = $x/10$

⇒ $x/10 = 12500$

⇒ $x = 125000$

Number of total people = 125000

The Number of people who casted votes = $4x/5$

= $(4 \times 125000)/5 = 1,00,000$

∴ **1,00,000 people casted their votes.**

Q. Amit got 60% of the votes in an election and he won by 930 votes. Find the number of total voters in the election.

Sol. Amit got 60% of the votes. Others got 40% of the votes.

The difference between them is $60 - 40 = 20\%$ Amit won by 930 votes = 20% votes

If $20\% = 930$ then, $100 \times 930 / 20 = 4650$

∴ **The total numbers of voters are 4650.**

★ Important Points

Consumption and Expenditure

If the price of a commodity increase by $R\%$, then the reduction in consumption so that the expenditure remains constant can be found by the formula:

$$\left[\frac{R}{(100 + R)} \times 100 \right] \%$$

Q. If the price of sugar is increased by 10%, then by how much per cent consumption should be reduced so that the expenditure remains the same?

Sol: By using formula:

$$\left[\frac{R}{(100 + R)} \times 100 \right] \%$$

Given: $R = 10\%$

$$\left[\frac{10}{(100 + 10)} \times 100 \right] \% = 9.09\%$$

📁 Alternate Method

Let the price be Rs. x /kg, Consumption be y kg

Hence, expenditure = price $\times$ consumption

⇒ Expenditure = xy

Price of sugar is increased by 10%

Hence, new price of sugar = $1.1x$ per kg

Let new consumption be z kg

Hence, new expenditure = $(1.1x) \times z$

Now, new expenditure = old expenditure

$$\Rightarrow (1.1x) \times z = x \times y$$

$$\Rightarrow z = y/1.1$$

Reduction in consumption = $(y - z) = y - (y/1.1) = y/11$

∴ **Percentage reduction in consumption = $[(y/11)/y] \times 100 = 100/11 = 9.09\%$**



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★ Important Points

Population-Based

Let the population of a town be P

Population after n years

Population after n years if rate of increment R is same

$$= P \left(1 + \frac{R}{100} \right)^n$$

Population after n years if rate of increment R is

$$\text{different} = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \dots$$

Q. The population of a town 2 years ago was 245000. It increased by 15% in the first year and then increased by 20% in the second year. What is the current population of the town?

Sol: The population of a town 2 years ago was 245000. It increased by 15% in the first year

∴ The population after the first year will be

$$= \frac{115}{100} \times 245000 = 281750$$

The population then increase by 20% in the second year.

∴ The population after the second year will be

$$= \frac{120}{100} \times 281750 = 338100$$

⚡ Shortcut Trick

Population n years Ago

$$\text{Population n years ago} = \frac{P}{\left(1 + \frac{R}{100} \right)^n}$$

Q. The current population of a town is 25710. If the current population of the town increases by 8% every year, what is the approximate population of the town 7 years ago?

Sol: Current population = 25710

$$\text{Population of town 7 years ago} = \frac{25710}{(1+0.08)^7}$$

$$= 25710/1.7138 = 15001.75 \approx 15000$$

★ Important Points

Depreciation-Based

Let the present value of a machine be P. Suppose it depreciates at the rate of R% per annum.

Value of the machine after n years

$$\text{Value of the machine after n years} = P \left(1 - \frac{R}{100} \right)^n$$

Q. An electric bulb was bought at Rs. 4100. Its value depreciates at the rate of 7% per annum. Its value after one year will be:

Sol: Actual price of the electric bulb = Rs. 4100

⇒ Depreciation rate = 7%

∴ Value after 1 year = 4100 – 7% of 4100

$$= 4100 - 4100 \times (7/100) = \text{Rs. } 3813$$

⚡ Shortcut Trick

$$P \left(1 - \frac{R}{100} \right)^n \Rightarrow 4100 \left(1 - \frac{7}{100} \right)^1 \Rightarrow 4100 \left(\frac{93}{100} \right) = \text{Rs. } 3813$$

∴ Value after 1 year = Rs. 3813



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★ Important Points

Fresh Fruit & Dry fruit Concept

Q. If fresh grapes contain 95% water, and dry grapes contain 5% water, how many kilograms of fresh grapes are required to get 400 gm of dry grapes?

Sol: Fresh grapes will have 5% pulp

Let the weight of fresh grapes be 'x' gms, then $x \times 5\% = 400 \times 95\%$

$$\Rightarrow x \times 5 = 400 \times 95$$

$$\Rightarrow x = 7600 \text{ gms}$$

∴ **The weight of fresh grapes is 7.6 kgs**

Q. Fresh fruit contains 65% water and dry fruit contains 25% water, then how much dry fruits can be obtained from 300 kg of fresh fruits (in kg)?

Sol: If 65% is water then 35% is pulp Fresh fruits $\times 35\% = \text{Dry fruits} \times 75\%$

$$\Rightarrow 300 \times 35 = \text{Dry fruits} \times 75 (\because \text{Fresh fruits} = 300 \text{ kg})$$

$$\Rightarrow \text{Dry fruits} = (300 \times 35)/75$$

∴ **Dry fruits = 140 kg**

★ Important Points

Tax Based

Total Tax Amount = $(100/\text{reduction}\%) \times \text{reduction amount}$

Tax amount paid = Total Tax amount - reduced amount

Q. If house tax is paid before the due date, one gets a reduction of 12% on the amount of the bill. By paying the tax before the due date, a person got a reduction of Rs. 2,100. The amount (in Rs.) of house tax paid was:

Sol: Total tax amount = $(100/12) \times 2100 = \text{Rs. } 17500$

Total amount paid = $17500 - 2100 = \text{Rs. } 15400$

∴ **The amount of house tax paid was Rs. 15400**

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Ratio and Proportion

Ratio: A ratio is a way of comparing quantities. It is a mathematical expression that represents the relationship between two numbers or quantities.

Ex:

The ratio of 4 to 2 can be represented as $4 : 2$ or $4/2$.

Proportion: A proportion is an equation that defines that two ratios are equal. It can be represented as $a : b = c : d$, where b and c are non-zero. Here, a and d are called extremes, where as b and c are called mean.

- In proportion, the product of the means is equal to the product of the extremes. (i.e., $ad = bc$).
- If $a : b = c : d$, then $a/c = b/d$. This is known as Componendo and dividendo rule.
- If $a : b = c : d$, then $(a + b)/b = (c + d)/d$. This is known as Componendo.
- If $a : b = c : d$, then $(a - b)/b = (c - d)/d$. This is known as dividendo.

★ Important Points

Types of ratio

Simple or Common Ratio: This is the most basic type of ratio, used to compare two quantities of the same kind.

Ex:

If there are 3 apples and 2 oranges, the ratio of apples to oranges is $3 : 2$.

Compound Ratio: This is essentially a ratio of ratios and is often used when more than two quantities are involved.

Ex:

If there are 3 apples, 2 oranges, and 1 pear, the compound ratio of apples to oranges to pears would be $3 : 2 : 1$.

Duplicate Ratio: This is the square of a simple ratio. If $a : b$ is a ratio, then $a^2 : b^2$ is the duplicate ratio.

Ex:

If the ratio of length to width of a rectangle is $2 : 3$, then the duplicate ratio is $4 : 9$

Triplicate Ratio: This is the cube of a simple ratio.

If $a : b$ is a ratio, then $a^3 : b^3$ is the triplicate ratio.

Ex:

If the ratio of width to height of a box is $1 : 2$, then the triplicate ratio is $1 : 8$.

Inverse or Reciprocal Ratio: This is when the two terms of a ratio are simply swapped. If $a : b$ is the original ratio, then $b : a$ is the inverse ratio.

Ex:

If the ratio of boys to girls in a class is $3 : 2$, then the inverse ratio, girls to boys, would be $2 : 3$.

Componendo Ratio: If $a : b = c : d$, then $(a + b) : b = (c + d) : d$ is the componendo ratio.

Dividendo Ratio: If $a : b = c : d$, then $(a - b) : b = (c - d) : d$ is the dividendo ratio.

Componendo and Dividendo Ratio: If $a : b = c : d$, then $(a + b) / (a - b) = (c + d) / (c - d)$ is the componendo and dividendo ratio.

★ Important Points

Types of Proportions

Third Proportion: In a proportion, if $a : b = b : c$, then 'c' is referred to as the third proportion of 'a' and 'b'. It means 'c' is the third number which maintains the balance of the proportion with the first two numbers.

Fourth Proportion: In a proportion, if $a : b = c : d$, then 'd' is referred to as the fourth proportion to 'a', 'b', and 'c'. It means 'd' is the fourth number which maintains the balance of the proportion with the first three numbers.

Mean Proportion: The mean proportion (or geometric mean) between two quantities a and b is the square root of the product of the quantities. So the mean proportion of 'a' and 'b' is $\sqrt{a \times b}$. It's a value that maintains the balance if you set a proportional comparison "a is to the mean proportion as the mean proportion is to b".

Q. Rs. 1800 is divided among A and B in the ratio of $2 : 3$ respectively. What is the share of A?

Sol. $\Rightarrow$ Total parts in the ratio = $2 + 3 = 5$

$\Rightarrow$ Each part = Total amount / Total parts in the ratio = $1800 / 5$

$\Rightarrow$ The share of A = $2 \times$ Each part

The share of A = Rs. 720

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Q. If three numbers are in the ratio 3 : 2 : 5 and sum of their squares is 1862. Then what will be the second number?

Sol. Let, the three number be 3m, 2m and 5m, respectively.

$$\Rightarrow (3m)^2 + (2m)^2 + (5m)^2 = 1862$$

$$\Rightarrow 9m^2 + 4m^2 + 25m^2 = 1862$$

$$\Rightarrow 38m^2 = 1862$$

$$\Rightarrow m^2 = 49$$

$$\Rightarrow m = 7$$

$$\Rightarrow \text{Second number} = 2m = 14$$

Q. If $x : y = 3 : 5$, then find the ratio of $(10x + 3y) : (5x + 2y)$.

Sol. $\Rightarrow x : y = 3 : 5$

Let 'a' be the common factor.

then, $x = 3a$ and $y = 5a$

$$\Rightarrow (10x + 3y) = 10(3a) + 3(5a) = 45a$$

$$\Rightarrow (5x + 2y) = 5(3a) + 2(5a) = 25a$$

$$\Rightarrow 45a : 25a = 9 : 5$$

★ Important Points

Problem on Age

Problems on ages commonly revolve around determining a person's present, past, or future age based on certain given conditions.

Q. The ratio of the present ages of Ajay and Rajveer is 7 : 4 respectively. After three years the ratio of their ages will be 5 : 3 respectively. Find the present age of Ajay.

Sol. Let, the ratio is 7x and 4x

$$\Rightarrow \text{Present age of Ajay} = 7x$$

$$\Rightarrow \text{Present age of Rajveer} = 4x$$

$$\Rightarrow \text{After 3 years, age of Ajay} = 7x + 3$$

$$\Rightarrow \text{After 3 years, age of Rajveer} = 4x + 3$$

$$\Rightarrow (7x + 3)/(4x + 3) = 5/3$$

$$\Rightarrow 3 \times (7x + 3) = 5 \times (4x + 3)$$

$$\Rightarrow 21x + 9 = 20x + 15$$

$$\Rightarrow x = 6$$

$$\text{Present age of Ajay} = 7 \times 6 = 42 \text{ years}$$

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RANK 1



ROHIT CHADHAR

SSC CHSL 2024
8 RANKERS IN TOP 20

RANK 1



MOHAN KUMAR

SSC JE 2024
5 RANKERS IN TOP 10

RANK 1



NIRANJAN JAIN

SBI PO 2024
5 RANKERS IN TOP 10

RANK 2



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Average

Average is defined as the sum of the observations divided by the number of observations.

$$\text{Average} = \frac{\text{Sum of the observations}}{\text{Number of observations}}$$

★ Important Points

Formula based on numbers:

Average of first n natural numbers

$$= (n + 1)/2$$

Average of squares of first n natural numbers

$$= (n + 1) (2n + 1)/6$$

Average of cubes of first n natural numbers

$$= n (n + 1)^2/4$$

Average of first n even numbers

$$= (\text{last even number} + 2)/2$$

Average of first n odd numbers

$$= (\text{last odd number} + 1)/2$$

Average of squares of first n even numbers

$$= 2n (n + 1) (2n + 1)/3$$

Average of squares first n odd numbers

$$= (2n + 1) (2n - 1)/3$$

Ex:

Q. Find the average of the first 101 natural numbers.

Sol: The average of first n natural number = $(n + 1)/2$

$$\Rightarrow \text{Average} = (101 + 1)/2$$

$$\Rightarrow \text{Average} = 102/2 = 51$$

$\therefore$ The average of first 101 natural numbers is 51.

Q. What is the average of squares of the first 10 natural numbers?

Sol: Sum of squares of first n natural Number

$$= [n (n + 1) (2n + 1)]/6$$

Sum of squares of first n natural Number

$$= [10(10 + 1)(2 \times 10 + 1)]/6 = 385$$

$\therefore$ Average of squares of first 10 natural numbers

$$= \text{sum of squares}/10 = 385/10 = 38.5$$

Q. The average of cubes of first 5 natural numbers is

Sol: Average of cubes of first n natural numbers

$$= n (n + 1)^2/4$$

Average of cube of first 5 natural numbers

$$= (25 \times 9)/5 = 45$$

Q. The average of all even numbers up to 86 is:

Sol:

Average of first n even numbers

$$= (\text{last even number} + 2)/2$$

$$\text{Average of first n even numbers} = (86 + 2)/2 = 44$$

Q. What is the average of the first 15 odd numbers among the natural numbers?

Sol: Total = Number of entities $\times$ Average

The sum of first n odd natural numbers = n^2

We know that, the sum of first n odd natural numbers = n^2

Then, the average of first n odd natural number

$$= n^2/n = n = 15$$

$\therefore$ The required average is 15



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Mixture and Alligation

In this chapter, we are going to learn a concept called Mixture and Alligation, which is used to solve various types of mixture problems.

There are two types of mixtures:

Simple Mixture:

This involves the mixing of two or more different entities.

Complex Mixture:

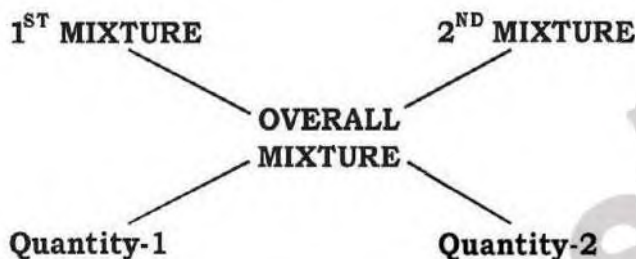
This involves the mixing of two or more different mixtures

Allegation:

'Allegation' is a method or rule that helps in solving problems related to these mixtures. This rule helps in finding the ratio in which two items must be mixed to obtain a final mixture with ingredients in a known ratio.

Ex:

It can be used to determine the mean value of the mixture when the proportions and amounts of the ingredients mixed vary



Important Points

When two mixtures are mixed to form a new mixture

Mean Price: This is the cost price of a unit quantity of the mixture.

Mean Price = (Quantity of cheaper × Price of cheaper + Quantity of dearer × Price of dearer) / (Quantity of cheaper + Quantity of dearer).

Quantity Ratio: The ratio of quantities of cheaper and dearer in the mixture.

Quantity Ratio = (CPD - Mean Price) / (Mean Price - CPC), where CPD is the cost price of the dearer item quantity and CPC is the cost price of the cheaper item quantity.

Removal and Replacement: If a quantity is removed and another quantity is added keeping the total volume the same, the formula used to calculate the final amount of original substance left after n operations:

Final Quantity

= Initial Quantity × (1 - Replacement Total Quantity)ⁿ

Ex:

Q. A bottle of capacity 80 litres contains alcohol and spirit in the ratio 7 : 3. How many spirits must be added to the mixture, so that the ratio of alcohol and spirit becomes 2 : 1?

Sol: Quantity of alcohol in bottle = $80 \times 7/10$
= 56 litres

Quantity of spirit in bottle = $80 - 56 = 24$ liters

Now, let 'x' litres of the spirit be added

$$\therefore 56/(24 + x) = 2/1$$

$$\Rightarrow 56 = 2 \times (24 + x)$$

$$\Rightarrow 56 = 48 + 2x$$

$$\Rightarrow x = 4$$

Q. A container contains 25 litre of milk. From this container, 5 litre of milk is taken out and replaced by water. This process is further repeated two times. How much milk is there in the container now?

Sol: Fraction of milk removed = $5/25 = 1/5$

Now, the quantity of milk left in the container

$$\Rightarrow 25(1 - 1/5)^3$$

$$\Rightarrow 25 \times (4/5)^3$$

$$\Rightarrow 25 \times 64/125$$

$$\Rightarrow 12.8 \text{ litre}$$



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Partnership Business

Partnership is based on three points, the first is time, the second is investment and the third one is profit. So, the ratio of profit is directly proportional to the product of investment and time.

i.e. Profit = Invested Capital $\times$ Invested Time

★ Important Points

Profit Distribution in Partnerships

(A) Different Investments, Same Time Period of Investing

If the amount invested by the partners are I_1, I_2, I_3 then the profit is distributed in the ratio $I_1 : I_2 : I_3$

(B) When the investment and the time period is different

Let there be three partners, one invests I_1 for t_1 time, second invests I_2 for t_2 time and third invests I_3 for t_3 time. The profit is shared in the ratio

$$I_1 \times t_1 : I_2 \times t_2 : I_3 \times t_3$$

(C) Different Amounts Invested In Different Time Periods

Let's say in a partnership between A and B, A invests Rs. I_a for a time period of t_a .

But B invests Rs. I_{b1} for a period of t_{b1} time and Rs.

I_{b2} for a period of t_{b2} time. In this case the profit will be divided between A and B in the ratio

$$I_a \times t_a : (I_{b1} \times t_{b1} + I_{b2} \times t_{b2})$$

Ex:

Q. P and Q invested in a business. The profit earned was divided in the ratio 2 : 3. If P invested Rs. 40000. What is the amount invested by Q?

Sol: The amount of P invested Rs. 40000 and amount invested by Q is y.

Amount invested P to Q = ratio of their profit

$$\Rightarrow 40000/y = 2/3$$

$$\Rightarrow y = 40000 \times 3/2 = \text{Rs. } 60000$$

$\therefore$ The amount invested by Q is Rs. 60000.

Q. A and B enter into a partnership. A contributed ₹ 5000 for 8 months and B ₹ 6000 for 5 months. Find A's share in a total profit of ₹ 8400.

Sol: A's and B's shares as product of invested amount and time:

$$\Rightarrow \text{A's share} = \text{Rs. } 5000 \times 8 = \text{Rs. } 40,000$$

$$\Rightarrow \text{B's share} = \text{Rs. } 6000 \times 5 = \text{Rs. } 30,000$$

Their share ratio is A : B = 40000 : 30000 = 4 : 3

Now, distribute total profit according to their shares:

$$\Rightarrow \text{A's profit} = (\text{A's share} / \text{total share}) \times \text{total}$$

$$= (4/7) \times \text{Rs. } 8400 = \text{Rs. } 4800.$$

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Profit and Loss & Discount

Definition

Cost Price (CP): It is the price that cost a manufacturer or person for a product or service.

Selling Price (SP): It is the price at which a product or service is sold to customers.

Profit (P): Whenever selling price is greater than cost price, the difference between is P.

$$\text{Profit} = \text{SP} - \text{CP}$$

Loss (L): Whenever selling price is Less than cost price, the difference between is L.

$$\text{Loss} = \text{CP} - \text{SP}$$

★ Important Points

1. Gain/Profit = Selling Price(S.P.) - Cost Price (C.P.)

2. Loss = Cost Price (C.P) - Selling Price(S.P)

3. If the rate of gain or loss are given then,

(a) Selling price =

$$\text{CP} \times \frac{100 - L\%}{100} \quad \text{or} \quad \text{CP} \times \frac{100 + P\%}{100}$$

$$(b) \text{ Cost Price} = \text{SP} \times \frac{100}{100 + P\%} \quad \text{or} \quad \text{SP} \times \frac{100}{100 - L\%}$$

(c) The gain or loss is always calculated on the cost price.

4. If $\text{SP} > \text{CP}$ then Gain = S.P - C.P

$$\text{Gain \%} = \frac{\text{gain} \times 100}{\text{CP}}$$

5. If $\text{SP} < \text{CP}$ then Loss = S.P - C.P

$$\text{Loss \%} = \frac{\text{Loss} \times 100}{\text{CP}}$$

$$6. \text{ In \% Gain Condition } \text{SP} = \text{CP} \times \frac{100 + P\%}{100}$$

$$7. \text{ If Loss \% Condition } \text{SP} = \text{CP} \times \frac{100 - L\%}{100}$$

$$8. \frac{\text{CP}}{\text{MP}} = \frac{100 - D\%}{100 \pm P/L\%}$$

9. If two items are sold at same price, each at Rs. x one at a profit of P% and other at a loss of P%, Then

$$\text{Loss} = \frac{P^2}{100} \%$$

10. If CP is same, Then change is SP % = Change in Profit/Loss %

11. If MP of two articles is same, then Difference between SP = Difference between discount.

Dishonest Dealings:

$$\text{Profit \%} = \frac{\text{Error}}{\text{True Weight} - \text{Error}} \times 100$$

Discount and Marked Price

Definition

Marked Price: It is the price at which a product or service is listed for.

Discount: It is the difference between marked price and selling price.

Important formula:

$$\text{Discount} = \text{MP} - \text{SP}$$

$$\text{Discount \%} = (D/\text{MP}) \times 100$$

$$\text{Selling price} = (100 - D\%) \times \text{MP}/100$$

Ex:

Q. Marked price of a cricket bat is Rs 1000 and it is sold at Rs 800. Find the discount percentage.

$$\text{Sol: Discount} = \text{MP} - \text{SP} = 1000 - 800 = \text{Rs } 200$$

$$\text{Discount Percentage} = (D/\text{MP}) \times 100 \\ = (200/1000) \times 100 = 20\%.$$

Q. Marked price of a product is Rs 240 and 25% of discount is provided on it. Find the selling price.

$$\text{Sol: Discount} = \text{SP} \times 25\% = 240 \times (25/100) = \text{Rs } 60$$

$$\text{Selling price} = \text{MP} - \text{Discount} = 240 - 60 = \text{Rs } 180.$$

Alternate Method

$$\text{Selling Price} = \text{MP} \times (100 - D\%)/100$$

$$= 240 \times (100 - 25)/100 = \text{Rs } 180.$$

Successive Discount

$$\text{Effective discount\%} = X + Y - \frac{XY}{100}$$

(where, X = 1st discount % and Y = 2nd discount %)

Ex:

Q. A T-shirt is sold after providing two successive discounts of 20%. If marked price of T-shirt is Rs 200 then find the selling price.

$$\text{Sol: Discount 1} = 200 \times 20/100 = \text{Rs } 40$$

$$\text{Selling price after 1st discount} = 200 - 40 = \text{Rs } 160$$

$$\text{Discount 2} = 160 \times 20/100 = \text{Rs } 32$$

$$\text{Selling price after 2nd discount} = 160 - 32 = \text{Rs } 128$$



★ Important Points

Dishonest Dealing

In this a person/shopkeeper sells any product at wrong weight and earn profit. This can be done either by using false weight or by false reading.

Ex:

1. A shopkeeper claims to sell rice at cost price but uses false weight of 900gm instead of 1000gm.
2. A person sells cloth to the customer but uses false reading and gives 90 m. cloth instead of 100 m.

$$\text{Profit \%} = \frac{\text{Error}}{\text{True Weight} - \text{Error}} \times 100$$

Ex:

Q. Using faulty weights, a dealer gives only 1,650 ml of milk to his customer instead of 2 liters. Find his profit percentage.

Sol: True weight = 2 liter = 2,000 ml

Error = 2,000 – 1,650 = 350 ml

$$\text{Profit \%} = \frac{\text{Error}}{(\text{True weight} - \text{Error})} \times 100 = \frac{350}{(2000 - 350)} \times 100 = 21.21\%$$

★ Important Points

Successive selling

In it a product is sold for more than one time from one person to another person at some profit or loss. For example - A sold a pen to B at 10% profit and then B sold the pen to C at 20% profit.

If there are two successive profits or losses at x% and y% respectively, then the resultant profit or loss % = $x + y + \frac{xy}{100}$.

For profit, we take x and/or y as +ve value.

For loss, we take x and/or y as –ve value.

Ex:

Q. If A bought an article at Rs.200 and sold it to B at 20% profit. Again B sold the article at 10% profit to C. Find the amount paid by C.

Sol: Price paid by B = $200 + (200/100 \times 20) = 200 + 40 = \text{Rs. } 240$

$\therefore$ Price paid by C = $240 + (240/100 \times 10) = 240 + 24 = \text{Rs. } 264$

📄 Alternate Method

Net profit = $20 + 10 + 20 \times 10/100 = 32\%$

📄 Additional Information

If two objects are sold at same selling price, one at x% profit and other at x% loss, then

$$\text{Loss} = \frac{X^2}{100}$$

If the cost price of x articles is equal to selling price of y articles, then

$$\text{Profit \%} = \frac{[X - Y]}{Y} \times 100$$

If ath part of items are sold at x% loss, then for making no profit no loss Required gain percentage in

$$\text{selling rest items} = \frac{ax}{1 - a}$$

Ex:

Q. A man sold 2 bicycles at same selling price. One at 20% loss and other at 20% profit. Find overall profit and loss percentage

Sol: Let selling price be 300x

Then, CP for 1st bicycle = 250x Then,

CP of 2nd bicycle = 375x

Hence, Net CP = 625x and net SP = 600x

Net loss % = $(25x/625x) \times 100 = 4\%$

📄 Alternate Method

We know, If two objects are sold at same selling price, one at x% profit and other at x% loss, then,

$$\text{Loss \%} = \frac{X^2}{100}$$

$$\text{Net loss \%} = \frac{20^2}{100} = 4\%$$

★ Important Points

Sales Tax

During purchasing any product we have to give certain tax to the government. This additional payment is known as sales tax. **Tax is always calculated on selling price of product.**

Ex:

Q. Ram purchased a bicycle for Rs. 5954 He had paid a VAT of 14.5%. Find the list price of the bicycle.

Sol: Let the list price be Rs. a. VAT

$$= 14.5\% \text{ So, } a \times (114.5/100) = 5954$$

$$\Rightarrow a = (5954 \times 100)/114.5 = \text{Rs. } 5200$$

$\therefore$ The list price of the bicycle was Rs. 5200.



Interest

It can be classified into two types:

1. Simple Interest
2. Compound Interest

Simple Interest

When the interest is calculated only on the Principal for every year, it is called Simple Interest.

$$SI = \frac{P \times R \times T}{100}$$

Where, P = Principal, R = Rate of interest per year, T = Time period in years.

Ex:

Q. Rs.1080 invested for 3 months gave an interest of Rs.27. The simple rate of interest per annum was:

Sol: 3 months = 3/12

$$SI = (P \times r \times t)/100$$

$$\Rightarrow 27 = (1080 \times r \times 3/12)/100$$

$$\Rightarrow 27 = (90 \times r \times 3)/100$$

$$\Rightarrow 27 = 270r/100$$

$$\therefore r = 10\%.$$

Compound Interest

It is the interest paid on the original principal amount and the accumulated past interest.

$$A = P\left(1 + \frac{R}{100}\right)^T$$

Where, A = Amount, P = Principal, R = Rate of interest, T = Time period

Amount = Principal + Compound Interest

Additional Information

Table for direct calculation in CI:

Rate	CI of 2 Years	CI of 3 years	CI of 4 years
5%	10.25%	15.7625%	21.5%
10%	21%	33.1%	46.41%
15%	32.25%	52.0875%	69.35%
20%	44%	72.8%	107.36%

Ex:

Q. What will be the amount of a sum of Rs.25000 is placed at CI for 3 years while the rate of interest for the first, second, and third years is 4%, 8%, and 10%, respectively?

Sol: $A = P(1 + r_1/100)(1 + r_2/100)(1 + r_3/100)$
 $= 25000(1 + 4/100)(1 + 8/100)(1 + 10/100)$
 $= 25000(104/100)(108/100)(110/100) = 30888$

Additional Information

- In CI Amount will double in "N" years then,

$$N = \frac{69}{r} + 0.35$$

- Difference between CI and SI for two year

$$(CI - SI)_{2\text{ year}} = P\left(\frac{R}{100}\right)^2$$

- Ratio of CI and SI

$$\left(\frac{CI}{SI}\right)_{\text{for 2 years}} = 1 + \frac{R}{200}$$

- Difference between CI and SI for three years

$$(CI - SI)_{3\text{ year}} = 3P\left(\frac{R}{100}\right)^2 + P\left(\frac{R}{100}\right)^3$$

Important Points

Tree-method

In this method we assume principal (on the basis of rate and time given) such that it eases our calculation part and at the last, we compare it to the value given in the question to get the required answer.

Ex:

If 10% interest rate is given for 2 years then we will assume the principal as Rs. 100 and if the time is 3 years then we will assume the principal as Rs. 1000. It is done to avoid any calculation in decimal form.

Q. Find compound interest for principle Rs 10000, time = 3 years, and rate = 10%.

Effective rate method

Sol: Effective rate for 3 years at rate of 10%

$$= 3a.3a^2a^3 = 33.1\%$$

$$\text{Hence compound interest} = 10000 \times 33.1\%$$

$$= \text{Rs. } 3310.$$

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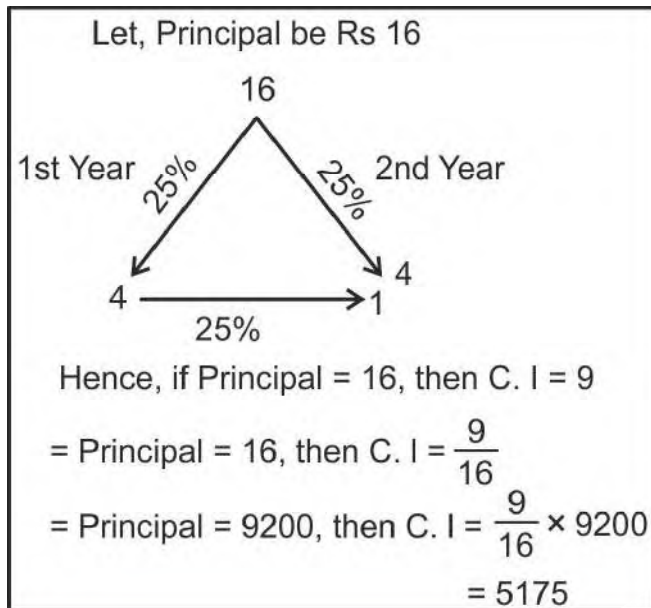


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Q. Rs.9200 is invested at compound interest at the rate of 25% per annum for 2

Sol:

A. Tree Method:



 **Alternate Method**

Effective Rate method:

Effective rate = $x + y + \frac{xy}{100} = 25 + 25 + \frac{(25 \times 25)}{100} = 56.25\%$. Hence, C.I = $9200 \times 56.25\% = \text{Rs } 5175$

Ex: Q. What will be the difference between the compound interest and simple interest on Rs.3000 at 10% rate of interest for 2 years?

$$D = PR^2/10^4$$

$$\Rightarrow D = (3000 \times 10^2)/10000$$

$$\therefore D = \text{Rs.}30$$

Q. What will be the difference between the compound interest and simple interest on Rs.100000 at 10% rate of interest for 3 years?

$$D = PR^2(R + 300)/10^6$$

$$D = 100000 \times 10^2 \times 10 + 300/10^6$$

$$\therefore D = \text{Rs. } 3100$$

Important Points

Installment

When the sum of an amount is paid in parts, then each part of the amount paid is called an installment.

For Simple Interest

Installment $(100 \times A)/[(100 \times t) + (r \times t \times (t - 1)/2)]$

Where A = Total amount paid, x = Value of each installment, r = Rate of interest

Ex:

Q. What annual installment will discharge a debt of Rs. 3,270 due in 3 years at 9% per annum simple interest?

A. Installment $(100 \times A)/[(100 \times t) + (r \times t \times (t - 1)/2)]$

$$\text{Installment} = (100 \times 3270)/[(100 \times 3) + (9 \times 3 \times (3 - 1)/2)]$$

$$\Rightarrow \text{Installment} = (100 \times 3270)/((300) + (27 \times 2/2))$$

$$\Rightarrow \text{Installment} = (100 \times 3270)/(300 + 27)$$

$$\Rightarrow \text{Installment} = 327000/327 = \text{Rs. } 1000$$

$$P = \left[\frac{x}{\left(1 + \frac{R}{100}\right)} + \frac{x}{\left(1 + \frac{R}{100}\right)^2} + \dots + \frac{x}{\left(1 + \frac{R}{100}\right)^n} \right]$$

$\therefore$ The annual installment will be Rs. 1000.

For Compound Interest

Q. If the Rs. 7500 taken at 8% compound interest is to be repaid in equal annual installment of 3 years, find the approximate annual installment.

Sol: Let the each installment be Rs. x

According to the question,

$$\Rightarrow 7500(1 + 8/100)^3 = x + x(1 + 8/100) + x(1 + 8/100)^2$$

$$\Rightarrow 7500 \times 1.083 = x + 1.08x + 1.664x$$

$$\Rightarrow 9447.84 = 3.2462x$$

$$\Rightarrow x = 9447.84/3.2464$$

$$\Rightarrow x = 2910(\text{Approx})$$

$\therefore$ The approximate annual installment is Rs. 2910.



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Speed Time and Distance

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Unit Conversions

km/hr to m/s

$$x \text{ km/hr} = \left(x \times \frac{5}{18} \right) \text{ m/sec}$$

Q. Convert 54 km/hr into m/sec.

A. 54 km/hr = $54 \times \frac{5}{18} = 15$ m/sec

★ Important Points

Speed $\propto$ 1/Time (When distance is constant)

Distance $\propto$ Time taken (when speed is constant)

Average Speed

Average speed is a measure of the distance travelled per unit of time

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total Time}}$$

Additional Information

$$\text{Average Speed} = (2 \times S_1 \times S_2) / (S_1 + S_2)$$

When distance is constant, average speed is free from distance.

Q. A truck covers a distance of 1200 km in 40 hours. What is the average speed of the truck?

Sol. A. Average speed = Total distance travelled / Total time taken

$$\Rightarrow \text{Average speed} = 1200/40$$

$$\therefore \text{Average speed} = 30 \text{ km/hr}$$

Q. A man travelled 12 km at a speed of 4 km/h and further 10 km at a speed of 5 km/hr. What was his average speed?

Sol.

Total time taken = Time taken at a speed of 4 km/h + Time taken at a speed of 5 km/h

$$\Rightarrow 12/4 + 10/5 = 5 \text{ hours } [\because \text{Time} = \text{Distance}/\text{Speed}]$$

Average speed = Total distance / Total time

$$\Rightarrow (12 + 10) / 5 = 22/5 = 4.4 \text{ km/h}$$

Q. Rahul goes Delhi to Pune at a speed of 50 km/h and comes back at a speed of 75 km/h. Find his average speed of the journey.

Sol.

As, distance is same both cases

$$\Rightarrow \text{Required average speed} = (2 \times 50 \times 75) / (50 + 75) = 7500/125 = 60 \text{ km/hr}$$

Relative Speed

It is defined as the rate at which two bodies are moving/separating from each other.

- When the two objects travel in the same direction their relative speed = $(a - b)$ km/hr
- When the two objects travel in the opposite direction their relative speed = $(a + b)$ km/hr

Where a and b are speeds of two objects in km/hr and $(a > b)$

Q. A thief is spotted by a policeman from a distance of 500 m. When the policeman started running after the thief, the thief also started running. Assuming the speed of the thief to be 20 km/hr and that of the policeman 25 km/hr, how far will the thief run before he is caught by the policeman?

Sol. Relative speed = $25 - 20 = 5$ km/hr [As they are running in the same direction]

$$\Rightarrow \text{Time taken to cover 500 m} = 500 / (5 \times \frac{5}{18}) \text{ sec}$$

$$\Rightarrow \text{Time} = 360 \text{ seconds} = 6 \text{ min} = 0.1 \text{ hr}$$

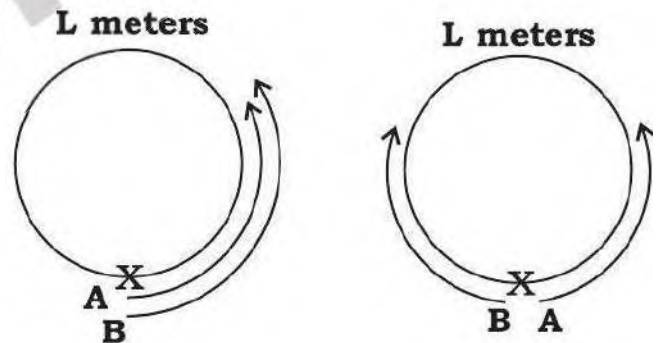
$$\Rightarrow \text{Distance covered by thief in 0.1 hr} = (20 \times 0.1) = 2 \text{ km}$$

Distance = initial distance $\times$ {speed of thief / (speed of policeman - speed of thief)}

$$\text{Distance} = 500/1000 \times \{20/(25 - 20)\} = 20/10 = 2 \text{ km}$$

★ Important Points

Circular track



Case 1: Opposite direction

Meet first time each other at any point on the track

$$\text{Time} = L/(x + y)$$

Case 2: Same direction

Meet first time each other at any point on the track

$$\text{Time} = L/(x - y)$$

Where x and y are speeds

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★ Important Points

Problems on Train

Train Crossing a Stationary Object (like a pole or a signal post):

When a train crosses a stationary object like a pole or a signal post, it needs to move a distance equal to its own length to completely pass the object.

Formula:

Time taken to cross = Length of Train / Speed of Train

Ex:

Q. Let's say there's a train 150 meters long traveling at a speed of 30 m/s. If this train is passing a stationary pole, how long will it take to completely cross the pole?

Sol. Using the formula:

Time taken = 150meters/ 30m/s = 5 seconds

So, the train takes 5 seconds to completely cross the pole.

Train Crossing a Stationary Extended Object (like a platform or a bridge):

If a train is crossing an extended stationary object, such as a platform or a bridge, then the distance the train needs to cover is the sum of the length of the train and the length of the object (platform/bridge).

Time taken to cross = (Length of Train + Length of Object) / Speed of Train

Ex:

Q. Consider a train that's 300 meters long traveling at a speed of 20 m/s. If this train is passing a stationary platform that's 200 meters long, how long will it take to completely cross the platform?

Sol.

Using the formula:

Time taken = (300 meters + 200 meters)/20m/s = 500 meters/20m/s = 25 seconds

So, the train takes 25 seconds to completely cross the platform.

Train passing a man/point object:

When a train passes a man/object, the distance travelled by the train while passing that object, will be equal to the length of the train.

Q. A train 320 meters long crosses a pole in 16 seconds. Find the speed of the train.

Sol.

As we know when a train crosses a pole it travels its own length.

Speed of train = 320/16 = 20 m/s

Q. Determine the length of train A if it crosses a pole at 60km/h in 30 sec.

Sol. Given, speed of the train = 60 km/h

⇒ Speed = 60 × 5/18 m/s = 50/3 m/s

Given, time taken by train A to cross the pole = 30 s

The distance covered in crossing the pole will be equal to the length of the train.

⇒ Distance = Speed × Time

⇒ Distance = 50/3 × 30 = 500 m

Train and Body Moving in the Same Direction:

If both the train and the moving object (e.g., man, car, another train) are moving in the same direction, the relative speed is the difference between their speeds.

Relative Speed = Speed of Train - Speed of Object

Time taken to cross = Length of Train/Relative Speed

Ex:

Q. A train 400 meters long is moving at 40 m/s. It is crossing a man running in the same direction at a speed of 5 m/s. How long will it take the train to cross the man?

Sol. Relative Speed = 40 m/s - 5 m/s = 35m/s

Time taken = 400 meters/35m/s = 11.43 seconds

Train and Body Moving in Opposite

Directions:

If the train and the moving object are moving towards each other, the relative speed is the sum of their speeds.

Relative Speed = Speed of Train + Speed of Object

Time taken to cross = Length of Train/Relative Speed

Ex:

Q. A train 500 meters long is moving at 50 m/s. It is crossing a man running towards the train at a speed of 10 m/s. How long will it take the train to cross the man?

Sol.

Relative Speed = 50 m/s + 10 m/s = 60 m/s

Time taken = 500 meters/60 m/s
= 8.33 seconds

Train Crossing Another Moving Train:

When two trains are crossing each other, the length of both trains should be considered when determining the distance covered during the crossing.

If both trains are moving in opposite directions:

Formula: Relative Speed = Speed of Train₁ + Speed of Train₂

Time taken to cross = (Length of Train₁ + Length of Train₂)/Relative Speed where the relative speed is the sum of the speeds of both trains

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★ Important Points

Boats and Streams

In such problems boat travels either in the direction of stream or in the opposite direction of stream. The direction of the boat along the stream is called down stream and the direction of boat against the stream is called upstream. If the speed of a boat in still water is u km/hr and the speed of the stream is v km/hr, then:
Speed downstream = $(u + v)$ km/hr
Speed upstream = $(u - v)$ km/hr

Q. A girl can row a certain distance downstream in 2 hours and upstream in 3 hours. If the stream flows at the rate of 4 km/h, what is the speed of the girl in still water?

Sol. Let, the distance = x km and speed of the girl in still water = y km/h Then,
 $\Rightarrow$ Downstream speed = $y + 4$ km/h
 $\Rightarrow$ Upstream speed = $y - 4$ km/h Now, according to the question, $x/(y - 4) = 3 \dots (1)$
And, $x/(y + 4) = 2 \dots (2)$
Solving equation (1) and (2) we get $y = 20$ km/h

Q. In 5 hours a boat can be rowed 16 km upstream or 28 km downstream. Find the speed of the boat in the still water.

Sol.

Speed = Distance/Time

Speed of downstream = $(28/5)$ km/hr = 5.6 km/hr

Speed of upstream = $(16/5)$ km/hr = 3.2 km/hr

Since, Speed of boat in still water = $\frac{1}{2}(\text{speed of down stream} + \text{speed of upstream})$

$\Rightarrow \frac{1}{2}(3.2 + 5.6)$

$\Rightarrow 4.4$ km/hr

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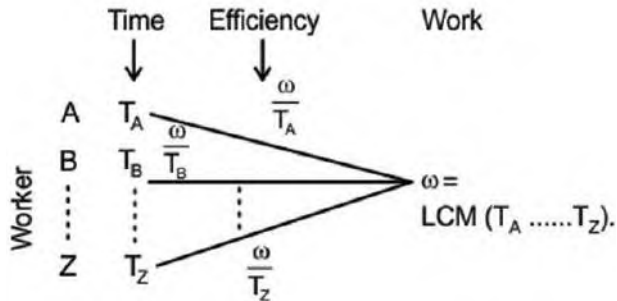
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Time and Work

Efficiency $\times$ Time = Total Work

★ Important Points

Efficiency is directly proportional to the work done and inversely proportional to the time taken.



When two persons do a work

Q. A does work in 10 days and B does the same work in 15 days. In how many days will they do the same work together?

Sol: A does work in 10 days

A's 1 day's work = $1/10$

B does work in 15 days

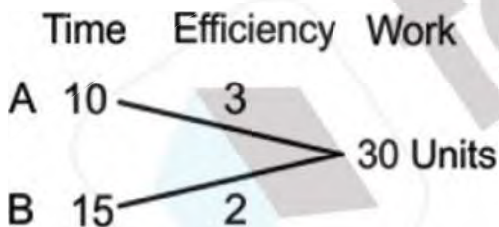
B's 1 day's work = $1/15$

Adding equation 1 and 2, we get,

A and B's 1 day's work = $1/10 + 1/15 = 5/30 = 1/6$

$\therefore$ A and B will together take 6 days to do the work.

Alternate Method



$A + B = 3 + 2 = 5$ units/day

Time = $30/5 = 6$ Days

When three persons do a work

Q. A and B can do a work in 3 days; B & C can do it in 4 days and A & C can do it in 6 days. In how many days will A, B & C finish it, if they work together?

Sol: Work done by A & B in 1 day = $1/3$... (I)

$\Rightarrow$ Work done by B & C in 1 day = $1/4$... (II)

$\Rightarrow$ Work done by A & C in 1 day = $1/6$... (III) Adding

(I), (II) & (III)

$\Rightarrow 2 \times$ (work done by A, B & C in 1 day)

= $1/3 + 1/4 + 1/6$

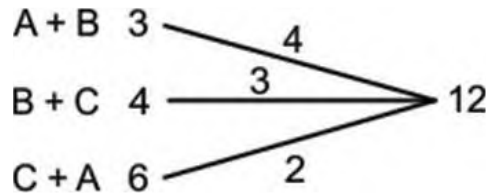
$\therefore$ Work done by A, B & C in 1 day together = $1/2$ ($1/3 + 1/4 + 1/6$)

= $1/2 [(4 + 3 + 2) / 12] = 3/8$

$\therefore$ No. of days taken to complete the job together

= $8/3$ days

Alternate Method



$A + B + C = 9/2$ units/days

Time taken = $12 / (9/2) = 24/9 = 8/3$ days

Questions based on Efficiency

Q. Anil is thrice as good a workman as Arun. Together they can do a job in 12 days. In how many days will Arun finish the work alone?

Sol: Anil is thrice as good a workman as Arun

If Anil can finish a work in x days, Arun can finish the same work in $3x$ days.

$\therefore$ 1 day's work by Anil = $1/x$ and 1 day's work by Arun = $1/3x$

Together they can do a job in 12 days

1 day's work by (Anil + Arun) = $1/x + 1/3x = 4/3x$

Now, $A/Q \ 4/3x = 1/12$

$\Rightarrow x = 16$

Time taken by Arun to finish the work alone = 48 days

★ Important Points

Work and wages

The ratio of wages of persons doing work is directly proportional to the ratio of efficiency of the persons.

Q. P, Q and R can complete work for Rs. 21000. P, Q and R can alone complete work in 20 days, 30 days and 60 days respectively. With the help of D, they complete the work in 8 days. Among the four who gets the highest share and how much?

Sol: P's 1 day's work = $1/20$

Q's 1 day's work = $1/30$

R's 1 day's work = $1/60$

(P + Q + R)'s 1 days work = $1/20 + 1/30 + 1/60$

= $(6/60) = 1/10$

(P + Q + R + D)'s 1 days work = $1/3$

D's 1 day's work = $1/8 - 1/10 = 2/80 = 1/40$

Ratio of work efficiency = P : Q : R : D

= $1/20 : 1/30 : 1/60 : 1/40 = 6 : 4 : 2 : 3$

P's share = $21000 \times 6/15 = \text{Rs. } 8400$

Q's share = $21000 \times 4/15 = \text{Rs. } 5600$

R's share = $21000 \times 2/15 = \text{Rs. } 2800$



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S' share = $21000 \times 3/15 = \text{Rs. } 4200$

P gets the highest share of Rs. 8400

When two or more persons work on alternate days or hours

Q. A and B can complete a piece of work in 10 days and 15 days respectively when working alone. Starting with A, they work on alternate days. In how many days will the work be completed?

Sol: Given,

A's 1 day's work = $1/10$

B's 1 day's work = $1/15$

(A + B)'s 2 days' work = $1/10 + 1/15 = 1/6$

To complete work (A + B) 12 days' work = $6 \times 1/6 = 1$

Total time taken by both if they work alternatively = 12 days.

MHDE/W based:

$$(M_1 \times D_1 \times T_1 \times E_1)/(W_1 \times C_1) = (M_2 \times D_2 \times T_2 \times E_2)/(W_2 \times C_2)$$

Where,

M is the number of working men,

D is the total number of days,

T is the total hours of working days,

E is the efficiency of working men,

W is the total work,

C is the consumption of working men

Q. 15 men can complete a task in 8 days. Three days after they started the work, 3 more men joined. In how many days will all of them together complete the remaining work?

Sol:

Total work done by 15 men in 8 days = $15 \times 8 = 120$

Work done by 15 men in 3 days = $15 \times 3 = 45$

Remaining work after 3 days = $120 - 45 = 75$

Days to complete remaining work by 18 men

= $75/18 = 25/6$ days

Food based

This type of question pertains to calculating the amount of food required to feed a certain group of people for a specific duration

Q. 1200 soldiers in a fort had enough food for 28 days. After 4 days, some soldiers left the fort. Thus, food lasted for 32 more days. How many soldiers left?

Sol:

Let the number of soldiers who left the fort be x. As we know,

$$M_1 \times D_1 = M_2 \times D_2 + M_3 \times D_3$$

$M_1 = 1200$ soldiers, $D_1 = 28$ days,

$M_2 = 1200$ soldiers, $D_2 = 4$ days,

$M_3 = (1200 - x)$ soldiers and $D_3 = 32$ days

$$M_1 \times D_1 = M_2 \times D_2 + M_3 \times D_3$$

$$1200 \times 28 = 1200 \times 4 + (1200 - x) \times 32$$

$$(1200 \times 28) - (1200 \times 4) = (1200 - x) \times 32$$

$$1200(28 - 4) = (1200 - x) \times 32$$

$$(1200 \times 24) / 32 = (1200 - x)$$

$$x = 300$$

Men, Women and Boys

Q. A man is twice as fast as a woman and a woman is twice as fast as a boy. If all of them i.e. a man, a woman and a boy can finish a work together in 4 days, in how many days a boy shall do it alone?

Sol:

Assume the efficiency of a boy = B units/day

As a woman is twice as fast as a boy so the efficiency of a woman = 2B units/day

As a man is twice as fast as a woman so the efficiency of a man = 4B units/day

According to the question, all of them i.e. a man, a woman and a boy can finish a work together in 4 days.

In 4 days total work done by a man, a woman and a boy = $4 \times (B + 2B + 4B) = 28B$ units.

Total work = 28B units.

Time required for a boy to do the whole work alone =

$$28B/B$$

Time required for a boy to do the whole work alone = 28 days.

★ Important Points

Pipe and Cistern

An "Inlet Pipe" fills the tank, and its rate is determined by the time it takes to fill the tank. An "Outlet Pipe" empties the tank, with a rate based on how quickly it can empty the tank. If both types of pipes are used together, their combined rate is the difference of their individual rates. The problems typically involve calculating how long it'll take to fill or empty a tank under various scenarios.

Ex:

Q. A pipe can normally fill a tank alone in 40 days but takes 20 days more due to a leak in the bottom of the tank. In how many days can the leak alone empty half the tank?

Sol:

$$\Rightarrow 1/40 + 1/B = 1/60$$

$$\Rightarrow 1/B = 1/60 - 1/40 \Rightarrow 1/B = -1/120$$

So leakage can empty the tank in = 120 days Since, half the tank is to be emptied so,

$$\Rightarrow \text{days required} = 120/2 = 60 \text{ days.}$$

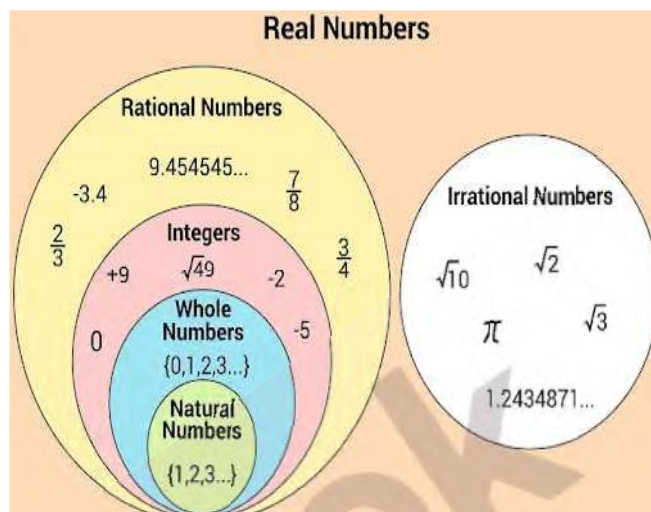


NUMBER SYSTEM

- 1. Natural Numbers:-** All counting numbers are called natural numbers and it is denoted by N. Ex: {1, 2, 3, 4...}
 - The smallest natural number is 1 but there is no largest natural number.
- 2. Whole Numbers:-** The set of all natural numbers together with zero are called whole numbers and it is denoted by W. Ex: {0, 1, 2, 3, 4...} Whole Number is also called as "Non-negative Integer"
- 3. Integers:-** The set of all natural numbers, zero and negative of all natural numbers are called integers and it is denoted by I and Z. Ex: {... -4, -3, -2, -1, 0, 1, 2 ...}
 - Negative integers are denoted by $I^- = \{-4, -3, -2, -1\}$
 - Positive integers are denoted by $I^+ = \{1, 2, 3, 4...\}$
 - '0' is a non-negative integer as well as a non-positive integer.
- 4. Rational Number:-** A number of the form p/q where p and q are whole numbers and q does not equal to zero and it is denoted by Q. Ex: $1/2, 2/3, -5/2, 0, 2, 5$
 - It can be negative, positive and also zero because we can write 0 as $0/1$.
- 5. Irrational Number:-** An irrational number is a real number that cannot be expressed as a ratio of integers. Ex: $\sqrt{2}, \sqrt{3}, \pi, e$ etc.
- 6. Real Number:-** The set of all rational numbers and irrational numbers are called Real Numbers. Ex: 1, $2/3, 0.6, \sqrt{3}, \pi, e$ etc.
 - The sum or difference between a rational and an irrational number is irrational. Ex: $(5 + \sqrt{2}), (5/2 - \sqrt{3})$ etc
 - The Product of a rational and an irrational number is irrational. Ex: $4\sqrt{2}, -5\sqrt{2}$ etc.
- 7. Complex Number:** A number of the form $(a + ib)$ is called a complex number where a and b are any real number and i is called iota ($\sqrt{-1}$). Ex: $\sqrt{-2} = 2i, 2 + \sqrt{-5} = 2 + i5$, etc.

★ Important Points

Some special numbers:-



- 1. Prime Number:-** A number is called the prime number that has only two factors (1 and itself). Ex: 2, 3, 5, 7, 11, 13... Etc.
 - 2 is the smallest even prime number
 - 1 is neither prime nor composite
- 2. Composite Number:-** Those numbers are called composite numbers which have more than two factors. Ex: 4 because the factor of 4 is 1, 2 and 4.
- 3. Co-Prime Number:-** A pair of any two integers is called a Co-Prime Number if their H.C.F is 1. Ex: (7, 11), (5, 11) etc.
- 4. Twin Prime Number:-** A pair of two prime numbers is called a twin prime number, if the difference between these numbers is 2. Ex: (3, 5), (5, 7), (11, 13)etc. Note:- (7, 11) is not a twin prime number.
- 5. Even and Odd Numbers:-** Integers which is divisible by 2 are called even number and which are not divisible by 2 is known as odd numbers. Ex: ..., -4, -2, 0, 2, 4,... etc. are even integers And ..., -3, -1, 1, 3, 5, ... etc. are odd integers

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Divisibility Rules

Rules by the help of which we can check the divisibility or factors of a number without actual division.

Rule #1:

Divisibility by 2: Numbers whose last digit is either even or zero are divisible by 2. **Ex:** 2, 28, 78, 10 etc.

Rule #2:

Divisibility by 3: If the sum of digits of a number is divisible by 3, then the number is also divisible by 3.
Ex: 72, 642, 96 etc.

Rule #3:

Divisibility by 4: If the last two digits of a Number are divisible by 4 or the number having two or more zeros at the end is divisible by 4.
Ex: 428, 772, 700 etc.

Rule #4:

Divisibility by 5: If the last digit of a number is divisible by 5 or 0, then the number is divisible by 5.
Ex: 50, 105, 100 etc.

Rule #5:

Divisibility by 6: If a number is divisible by 2 and 3, then the number is also divisible by 6.
Ex: 954, 732 etc.

Rule #6:

Divisibility by 7: If the difference between twice the unit digit of the given number and the remaining part of the given number is a multiple of 7 or it is equal to 0, the whole number is divisible by 7.
Ex: 322, 805 etc.

Rule #7:

Divisibility by 8: If the last three digits of a number are divisible by 8 or the last three digits of a number are zeros, then the number is divisible by 8.
Ex: 8432, 5000 etc.

Rule #8:

Divisibility by 9: If the sum of all the digits of a number is divisible by 9, then the number is also divisible by 9.
Ex: 8946, 765 etc.

Rule #9:

Divisibility by 11: If the difference between the Sum of digits at an even place and the Sum of digits at the odd place is divisible by 11, then the whole number is divisible by 11

Ex; 54659, 968 etc.

Rule #10:

Divisibility by 12: The number which is divisible by both 3 and 4 is also divisible by 12.

Ex: 456, 588 etc.

Unit Digit

Unit digit is used to represent the "ones" place value of any number or product.

Ex: $123 \times 256 = 31488$, '8' is the unit digit in the given product. In competitive examination we get different types of questions of Unit digits like

- Find the unit digit of $(342)^{43}$
- Find the unit digit of $(4567)^{987} \times (3846)^{84} \dots$ and so on.

Additional Information

Finding unit digit of an exponential Number

Cyclicity of Numbers

Num-ber (n)	Unit digit of n^1	Unit digit of n^2	Unit digit of n^3	Unit digit of n^4	Cy-clicity
0	0	0	0	0	1
1	1	1	1	1	1
2	2	4	8	6	4
3	3	9	7	1	4
4	4	6	4	6	2
5	5	5	5	5	1
6	6	6	6	6	1
7	7	9	3	1	4
8	8	4	2	6	4
9	9	1	9	1	2

First of all divide the power of that number by 4.

Condition (i): If you get the remainder zero then changes the power of that number by 4



Ex: We have to find the unit digit of this number $(3846)^{84}$

Step 1st: At first, we have to divide 84 by 4

Step 2nd: After dividing you will get 0 as remainder

Step 3rd: Change the powers of a given number with 4 and then find the unit digit by multiplying the unit digit four times

In the above condition $(3846)^{84} = (3846)$

The unit digit will be the unit digit of $6 \times 6 \times 6 \times 6$.

That is 6

Condition (ii): If you get the remainder other than zero i.e. (1, 2, 3)

Step 1st: Change the power of the given number with the remainder that you get in the above divisions.

Step 2nd: Then multiply the unit digit of that number in the times equal to the remainder.

Ex: If we have to find the unit digit of $(342)^{43}$
After dividing 43 by 4, we will get 3 as the remainder.

So, we can write $(342)^{43}$ as $(342)^3$ to find the unit digit. Now after multiplying 2 three times,

we will get the unit digit of $(342)^{43}$

number = $2 \times 2 \times 2 = 8$ is the unit digit.

Important Points

Trailing Zeroes

A **trailing zero** is the representation of a number that has no non-zero digits to the right of it. Let's take the number 705032010000 and try to find out the number of zeroes. The number 705032010000 has 4 trailing zeroes.

Here, we are counting the last 4 zeroes and the remaining 3 zeroes will not be counted as there are other non-zero digits between them.

Number of zeros at the end of the number can be found by finding the number of pairs of 2 and 5.

Q. Find the number of trailing zero in 234!

Sol: To find the number of zeroes at the end of the product, we need to calculate the number of 2's and number 5's or the number of pairs of 2 and 5.

$$\Rightarrow 2 \times 5 = 10$$

Number of zeroes = 1 (number of pair = 1)

The number of pairs of 2 and 5 is the same as the number of zeroes at the end of the product

To calculate the number of 2's and number of 5's in any factorial value or a series starts from

1 is to divide the last number by 2 or 5 (successive quotient) till 0 is the last quotient

According to the question,

$$\text{Quotient of } 234/5 = 46$$

$$\text{Quotient of } 46/5 = 9$$

$$\text{Quotient of } 9/5 = 1$$

$$\therefore \text{Total number of zeros} = 46 + 9 + 1 = 56$$

Q. Find the number of zeroes in

$$10 \times 20 \times 30 \times \dots \times 1000.$$

$$\text{Sol.} \Rightarrow (10 \times 1) \times (10 \times 2) \times (10 \times 3) \times (10 \times 4) \dots \times (10 \times 100)$$

$$\Rightarrow 10^{100} \times (1 \times 2 \times 3 \times \dots \times 100)$$

$$\Rightarrow 10^{100} \times (100!)$$

$$\text{Number of zeroes} = 100 + \{(100)/5 + (20)/5\}$$

$$\Rightarrow 100 + 20 + 4$$

$$\Rightarrow 124$$

Important Points

Divisibility and Remainder

Dividend = Divisor $\times$ Quotient + Remainder

When any positive number A is divided by any other positive number B and if $B > A$, then the remainder will be A itself. In other words, if the numerator is smaller than the denominator, then the numerator will be the remainder.

Ex: Remainder of $5/12 = 5$,

Remainder of $21/45 = 21$ etc.

The remainder should always be calculated in its actual form i.e. you cannot reduce the fraction to its lower ratio.

Ex:

The remainder of $1/2 = 1$,

The remainder of $2/4 = 2$,

The remainder of $3/6 = 3$ etc.

It can be observed that despite all the fraction being equal, the remainders are different in each case.



★ Important Points

Theorem Method

The product of any two or more than two natural numbers has the same remainder when divided by any natural number, as the product of their remainders.

Ex:

Remainder of $(12 \times 13)/7$
= Remainder of $156/7 = 2$

Theorem method:

So, first of all, we will find out the remainder of each individual number and then we will multiply these individual remainders to find out final remainders.

Since, Remainder of $12/7 = 5$

and Remainder of $13/7 = 6$

So, Remainder of $(12 \times 13)/7$

= Remainder of $(5 \times 6)/7$

= Remainder of $30/7 = 2$

- If $\frac{a^x}{a-1}$ then the remainder will always be 1, whether x is even or odd.

- If $\frac{a^{(P-1)}}{P}$ the remainder will be 1, where P is a prime number.

- If $\frac{(ma-1)^x}{a}$ then remainder will be (a-1), where x is an odd number.

- If $\frac{(ma+1)^x}{a}$ then remainder will be 1, where x is an even number.

- When $(p-1)^n$ is divided by 'p', the remainder will be '1' only if 'n' is an even number

Q. What is the remainder after dividing the number 37^{1000} by 9?

Sol. $(37)^{1000} = (36 + 1)^{100}$ Here, 36 is divisible by 9,
Since 36 is divisible by 9
Remainder from 36 = 0

Now, (1^{1000}) divided by 9 will always give you the remainder 1 Hence, remainder is 1

🔑 Shortcut Trick

Q. If 13^{50} is divided by 14, the remainder is

Sol: 13 can be written as $(14 - 1)$

13^{50} is divided by 14 $(14 - 1)^{50}/14$

50 is an even number. Remainder = 1

Q. When 78654 is divided by 9, the remainder is A.

Sol. $78654 = 9 \times 8739 + 3$ On dividing by 9,
Remainder = 3

Apply Digital Root method which is the sum of digits when divided by 9 will give the same remainder when the number is divided by 9.

The sum of digits = $7 + 8 + 6 + 5 + 4 = 30$

When 30 is divided by 9, will give the remainder as 3
Remainder = 3

★ Important Points

Number of Factors

To find the factors of any number, the number of factors, the sum of all factors, the number of even factors, the sum of all even factors, the number of odd factors and the sum of odd factors, we can follow these rules and formulae (Let the number be 480)

Rule: First of all find all the prime factor of the given number by factorization method

2	480
2	240
2	120
2	60
2	30
3	15
	5

So, $480 = 2^5 \times 3 \times 5$

- **Number of factors** = Multiplication of (powers of each prime factor + 1)
Number of factors = $(5 + 1) \times (1 + 1) \times (1 + 1) = 24$
- **Sum of all factors** = $(a^0 + a^1 + \dots + a^m) \times (b^0 + b^1 + \dots + b^n) \times (c^0 + c^1 + \dots + c^q)$, where a, b and c are the factors and m, n and q are the powers of the prime factor respectively.
So, Sum of all factors = $(2^0 + 2^1 + \dots + 2^5) \times (3^0 + 3^1) \times (5^0 + 5^1) = 1512$
- **Number of even factors** = Multiplication of (powers of each prime factor + 1, except



even number So, Number of even factors

$$= 5 \times (1 + 1) \times (1 + 1) = 20$$

- **Sum of even factors** = $(a^1 + \dots + a^m) \times (b^0 + b^1 + \dots + b^n) \times (c^0 + c^1 + \dots + c^q)$, where a (even prime factor), b and c are prime factors and m , n and q are the powers of the prime factor respectively.

$$\text{So, Sum of even factors} = (2^1 + \dots + 2^5) \times (3^0 + 3^1) \times (5^0 + 5^1) = 1488$$

- **Number of odd factors** = multiplication of (powers + 1) of odd prime factors

$$\text{So, Number of odd factors} = (1 + 1) \times (1 + 1) = 4$$

- **Sum of odd factors** = $(b^0 + b^1 + \dots + b^n) \times (c^0 + c^1 + \dots + c^q)$... Where b and c are odd prime factors, and n and q are the powers of the odd prime factors respectively.

$$\text{So, Sum of odd factors} = (3^0 + 3^1) \times (5^0 + 5^1) = 24$$

- **Number of prime factors** = Sum of powers of prime factors

$$\text{So, Number of prime factors} = 5 + 1 + 1 = 7$$

Q. Find the number of factors and addition of all the factors for 950.

Sol. Number of factors,

$$950 = 2 \times 5 \times 5 \times 19 = 2^1 \times 5^2 \times 19^2$$

$$\text{The number of factors} = (1 + 1) \times (2 + 1) \times (1 + 1)$$

$$= 2 \times 3 \times 2 = 12$$

Addition of all the factors

$$= (20 + 21) \times (20 + 21 + 22) \times (20 + 21)$$

Addition of all factors,

$$= (1 + 2) \times (1 + 2 + 4) \times (1 + 2)$$

Addition of all factors,

$$= 3 \times 7 \times 3 = 54$$

★ Important Points

LCM and HCF

Highest Common Factor (H.C.F): When the greatest number divides perfectly two or more numbers then that number is called the H.C.F of these numbers and it is called highest common factor.

We can find the H.C.F of two or more numbers by division method and factorization method

Formula: H.C.F of two or more numbers = product of common prime factors of lowest power

Ex: H.C.F (24, 288)

$$24 = 2^3 \times 3 \text{ and } 288 = 2^5 \times 3^2$$

$$\text{So, H.C.F} = 2^3 \times 3 = 24$$

Lowest Common Multiple (L.C.M): The least number which is divisible by two or more given numbers, that least number is called L.C.M. of the numbers.

Formula: L.C.M. of two or more numbers = product of highest power of each prime factors

$$\text{L.C.M. (24, 288)}$$

$$24 = 2^3 \times 3 \text{ and } 288 = 2^5 \times 3^2$$

$$\text{So, the L.C.M of (24, 288)} = 2^5 \times 3^2 = 288$$

L.C.M of fractions = L.C.M of numerators/ H.C.F of denominators

H.C.F of fractions = H.C.F of numerators/ L.C.M of denominators

Ex: L.C.M of $5/4, 8/3, 7/2$ = L.C.M of (5, 8, 7)/ H.C.F of (4, 3, 2) So, $280/1 = 280$

Ex: H.C.F of $5/4, 8/3, 7/2$ = H.C.F of (5, 8, 7)/ L.C.M of (4, 3, 2) So, $1/12$

Additional Information

When a number is divided by a , b or c leaving same remainder ' r ' in each case then that number must be $k + r$ where k is L.C.M of a , b and c .

- When a number is divided by a , b or c leaving remainders p , q or r respectively such that the difference between divisor and remainder in each case is same i.e., $(a - p) = (b - q) = (c - r) = t$ (say), then that (least) number must be in the form of $(k - t)$, where k is LCM of a , b and c
- The largest number which when divide the numbers a , b and c the remainders are same then that largest number is given by H.C.F. of $(a - b)$, $(b - c)$ and $(c - a)$.
- The largest number which when dividing the numbers a , b and c give remainders as p , q , and r respectively is given by H.C.F. of $(a - p)$, $(b - q)$ and $(c - r)$.
- Greatest n -digit number which when divided by three numbers p , q , r leaves no remainder will be Required Number = $(n - \text{digit greatest number}) - R$, R is the remainder obtained on dividing the greatest n -digit number by L.C.M of p , q , r .
- The n -digit largest number which when divided by p , q , r leaves remainder ' a ' will be the required number = $[n - \text{digit largest number} - R] + a$ where, R is the remainder obtained when $n - \text{digit largest number}$ is divided by the L.C.M of p , q , r .



★ Important Points

SURDS AND INDICES

Indices: Power is greater than 1

Surd: Power is less than 1

Ex. $5^{7/3}$, Hence 7 is indices and $1/3$ is surds

Laws of Indices

$$a^m \times a^n = a^{(m+n)}$$

$$a^m \div a^n = a^{(m-n)}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n$$

$$1/a^n = a^{(-n)}$$

$$(a/b)^n = a^n/b^n = a^n \div b^n$$

$$a^{(-n)}/b^{(-m)} = b^m/a^n$$

$$a^{(m/n)} = \sqrt[n]{a^m}$$

$$a^0 = 1 (a \neq 0)$$

$$a^1 = a$$

$$a^{(p/q)} = (a^p)^{(1/q)}$$

$$a^x = a^y \Rightarrow x = y$$

$$a^x = b^y \Rightarrow a = b$$

★ Important Points

Simplification

BODMAS:

BODMAS is a mnemonic acronym widely used in mathematics to help people remember the order in which calculations must be done within complex mathematical equations.

Q. Find the value of $56 \div 8 \times 3 + 30 - 25 \div 5 + 10$.

Sol: $56 \div 8 \times 3 + 30 - 25 \div 5 + 10$

$$\Rightarrow 21 + 30 - 5 + 10$$

$$\Rightarrow 21 + 35$$

$$\Rightarrow 56$$

★ Important Points

Decimal

Terminating Decimal:

A terminating decimal is a decimal number that ends after a certain number of decimal places. It does not continue indefinitely.

Ex:

The fraction $1/4$ in decimal form is 0.25. It stops at two decimal places, hence it is a terminating decimal.

Non-Terminating Decimal:

A non-terminating decimal is a decimal number that continues infinitely and doesn't stop.

Types:

a) **Repeating Decimal;** A non-terminating decimal where a pattern of one or more digits repeats indefinitely.

Ex:

The fraction $1/3 = 0.333...$, the digit 3 repeats indefinitely, hence it's a non-terminating repeating decimal.

b) **Non-Repeating Decimal:** A non-terminating decimal in which the digits do not repeat in a pattern. These are typically represented by irrational numbers.

Ex:

The value of π (Pi) is approximately 3.1415926535..., the digits after the decimal point continue infinitely without a specific pattern, hence this is a non-terminating non-repeating decimal.

Q. If $x = 0.05 \times 0.36 \div 0.4 + 0.055 + 1.50 \div 0.03$, what is the value of x ?

Sol. $x = 0.05 \times 0.36 \div 0.4 + 0.055 + 1.50 \div 0.03$

$$\Rightarrow x = 0.05 \times 0.9 + 0.055 + 50$$

$$\Rightarrow x = 0.045 + 0.055 + 50$$

$$\Rightarrow x = 50.1$$

Q. Solve: $5 + 0.5 + 6 + 0.5 + 7 + 0.5 + 0.5 = ?$

Sol. $\Rightarrow (5 + 6 + 7) + (0.5 + 0.5 + 0.5 + 0.5)$

$$\Rightarrow (18 + 2) = 20$$

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SERIES

Progression

An arrangement of numbers in definite order is called "Progression".

Progression can be classified into three types:

1. Arithmetic Progression
2. Geometric Progression
3. Harmonic Progression

Arithmetic Progression

- An arithmetic progression is a sequence of numbers in which each term is obtained by adding a fixed number to the preceding term except the first term (a_1).
- This fixed number is known as the **common difference (d)** of the A.P and it can be positive, negative or zero.
- If the first term of an A.P. is a_1 , the second term is a_2 , n^{th} term a_n and the common difference is d . Then the A.P. becomes a_1, a_2, a_3, a_n . Common difference (d):

$$a_2 - a_1 = a_3 - a_2 \dots \dots \dots a_n - a_{n-1} = d$$

Ex: 10, 12, 14, 16

Standard Form of Arithmetic Progression

$a, a + d, a + 2d, a + 3d, a + (n - 1) d$

Where,

a is the first term

d is common difference n is the number of terms

n^{th} term (a_n) = $a + (n - 1) d$. Where,

a is the first term

n is the number of terms

d is the common difference

Ex:

Q. Find the 20th term of the A.P: 20, 24, 28, 32

Sol. Sequence 20, 24, 28, 32

Here, First term (a_1) = 20

Common difference (d) = $24 - 20 = 4$

Number of term (n) = 20

n^{th} term of A.P = $a + (n - 1) d$

$$\Rightarrow 20 + (20 - 1) 4$$

$$\Rightarrow 20 + (19) \times 4$$

$$\Rightarrow 96$$

$\therefore$ 20th term of the above series is 96

Additional Information

Sum of first n term of Arithmetic Progression

$$S = \frac{n}{2} [2a + (n - 1)d]$$

Where, a is the first term, n is the number of terms, d is common difference

Sum of first n term of Arithmetic progression when first term and last term are given.

$$S = \frac{n}{2} (a + l)$$

Where a is the First term, n is the number of terms, and d is a common difference

Number of terms, when first, last term and common difference is given.

$$n = \frac{(l - a)}{d} + 1$$

Important Points

Arithmetic Mean

If a, b, c are in AP, then $b = (a + c)/2$

b is called the arithmetic mean of a and c

Geometric Progression

GP is a sequence of the number in each term is obtained by multiplying a fixed number to the preceding term except the first term.

This fixed number is known as the **common ratio (r)** of the G.P and it can be positive and negative.

If the first term of a G.P. is a_1 , the second term is $a_2, \dots, n^{\text{th}}$ term is a_n and the common ratio is r . Then the G.P. becomes $a_1, a_2, a_3, \dots, a_n$.

Common ratio (r): $a_2/a_1 = a_3/a_2 \dots = a_n/a_{n-1} = r$

Ex:

(i) 4, 12, 36, 108

Standard Form of Geometric Progression

$a, ar, ar^2, ar^3, \dots, ar^{(n-1)}$

Where, a is the first term, r is the common ratio, n is the number of terms

n^{th} term of Geometric Progression

Let Geometric progression $a_1, a_2, a_3, \dots, a_n$

n^{th} term (a_n) = $ar^{(n-1)}$

Where, a is the first term, n is the number of terms r is the common ratio.



Ex:

Q. Find the 15th term of the G.P: 1200, 600, 300,.....6th term

Sol. Sequence 1200, 600, 300, 6th term

First term (a) = 1200

Common ratio (r) = 600/1200 = 1/2

Number of terms = 6

nth term (a_n) = $ar^{(n-1)}$

$\Rightarrow 1200 \times (1/2)^{(6-1)}$

$\Rightarrow 1200 \times (1/2)^5$

$\Rightarrow 1200/32$

$\Rightarrow 37.5$

$\therefore$ 15th term of GP is 37.5

Additional Information

Sum of first n term of Geometric

Progression (When $r > 1$)

$$S_n = a \left(\frac{r^n - 1}{r - 1} \right)$$

Where, a is the first term, n is the number of terms r is common ratio

Sum of infinite terms of Geometric Progression (When $r < 1$)

$$S_{\infty} = a \left(\frac{1 - r^n}{1 - r} \right)$$

Where, S_{∞} is Sum of infinite terms of Geometric Progression a is First term, r is common ratio

Important Points

Geometric Mean

- If a, b, c are in G.P., then $b = \sqrt{ac}$
- b is called the Geometric mean of a and c

Harmonic Progression

Harmonic progression is a sequence of numbers in which reciprocal of each term in the sequence are in arithmetic progression.

General form of an H.P

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots$$

Where, a is the first term, d is common difference

Formula derivation of an H.P

Let the $a_1, a_2, a_3, \dots, a_n$ is an A.P. whose common difference is d.

Then, $a_1 = a$

$a_2 = a + d$

$a_3 = a + 2d$

$a_4 = a + 3d$

And, so on..... $a_n = a + (n-1)d$

Additional Information

If $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots$ are in H.P

Then nth term of HP = $\frac{1}{a + (n-1)d}$

Important Points

Harmonic Mean

If a, b, c are in H.P., then $b = 2ac/(a+c)$

b is called the Harmonic mean of a and c



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MENSURATION

Mensuration is mainly divided into three parts: One is **Area**, the second is **Volume** and the third is **Perimeter**.

Perimeter: The perimeter of a shape is the total distance around its outermost edges. The units of the perimeter are typically length units such as meters, centimetres, or feet.

Area: The area of a shape is the amount of space it occupies in a two-dimensional plane. It's usually measured in square units, such as square inches, square feet, square centimetres, square meters, etc.

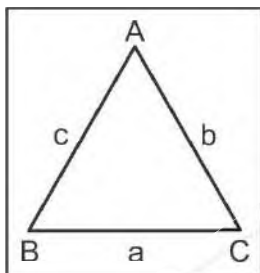
Volume: Volume is a measure of the amount of space an object or substance occupies in three dimensions. It's usually measured in cubic units like cubic inches, cubic feet, cubic centimetres, cubic meters, etc.

Mensuration 2D

Triangle

Scalene Triangle

A scalene triangle is a type of triangle in which all three sides have different lengths.



Let ABC be a triangle and a, b, c be its sides then

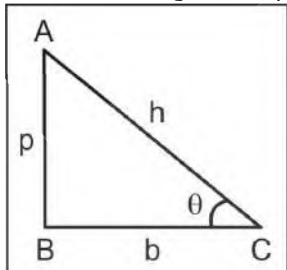
(i) Perimeter = $a + b + c$

(ii) Semi-perimeter = $s = (a + b + c)/2$

(iii) Area = $\sqrt{s(s-a)(s-b)(s-c)}$

Right Angle Triangle

A right-angled triangle or a right triangle is a type of triangle that has one angle exactly equal to 90 degrees.



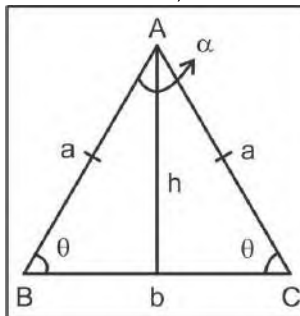
(i) Hypotenuse² = (base)² + (height)²

(ii) Perimeter = $p + b + h$

(iii) Area = $\frac{1}{2} \times \text{base} \times \text{height}$

Isosceles Triangle

An isosceles triangle is a triangle that has two sides of equal length. If ABC be an isosceles triangle such that $AB = AC = a$ and $BC = b$ then,



(i) $AB = AC = a$

(ii) $\angle B = \angle C$

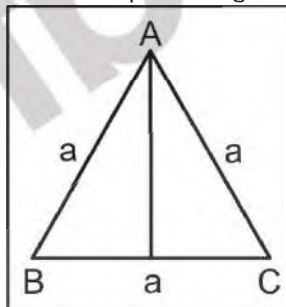
(iii) Perimeter = $2a + b$

(iv) Area = $\frac{b}{4} \sqrt{4a^2 - b^2}$

(v) $h = \sqrt{a^2 - \frac{b^2}{4}}$

Equilateral Triangle

An equilateral triangle is a type of triangle in which all three sides are equal in length.



(i) Perimeter = $3a$

(ii) Height (h) = $(\sqrt{3}a)/2$

(iii) Area = $\frac{\sqrt{3}a^2}{4}$

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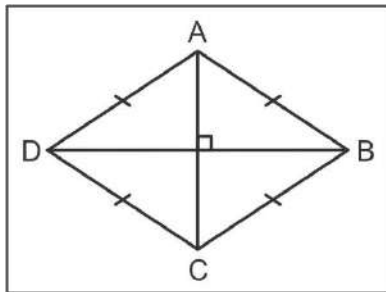
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Rhombus



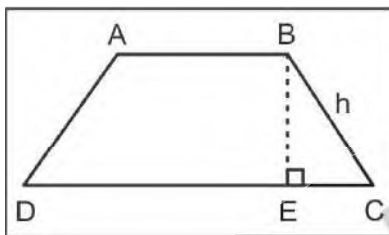
Important Points

- Area of rhombus = $\frac{1}{2} \times$ product of the lengths of its diagonals = $\frac{1}{2} \times d_1 \times d_2$
- d_1 and d_2 are diagonals, Side = $\frac{1}{2} \times \sqrt{d_1^2 + d_2^2}$

Additional Information

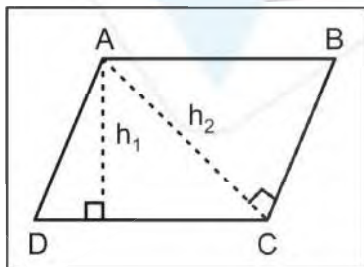
- A quadrilateral having all sides equal and their diagonals intersect at 90° .
- $AB = BC = CD = DA$

Trapezium



- A quadrilateral having a pair of sides is parallel and unequal. $AB \parallel DC$
- Area of trapezium ABCD = $\frac{1}{2} \times$ (sum of its parallel sides) $\times$ height
= $\frac{1}{2} \times (AB + DC) \times h$

Parallelogram

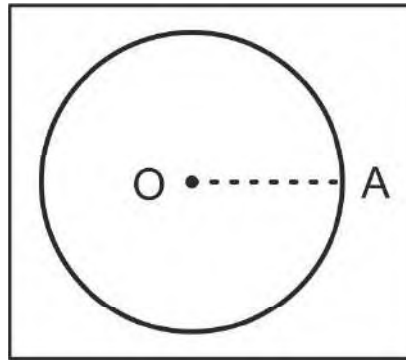


- Area of parallelogram ABCD = Base $\times$ Altitude on corresponding base = $DC \times h_1 = BC \times h_2$

Additional Information

- A quadrilateral having opposite sides are parallel and equal.
- $AB = DC$ and $AD = BC$
- $AB \parallel DC$ and $AD \parallel BC$

Circle



Important Points

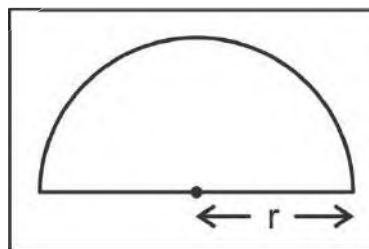
- Diameter = $2 \times$ radius
- Area = πr^2
- Perimeter = $2\pi r = \pi d$
- Length of arc = $\frac{\theta}{360^\circ} \times 2\pi r$
- Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

Additional Information

- A circle is a round, two-dimensional shape. All points on the edge of the circle are at the same distance from the center.
- O is the centre of this circle and OA is the radius.

Semi-circle

- Area of semi-circle = $\frac{1}{2} \pi r^2$
- Perimeter of semi-circle = $\pi r + 2r$



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Mensuration 3D

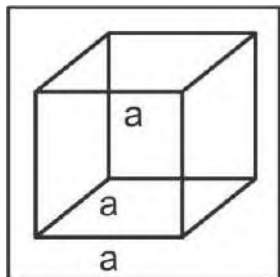
This topic contains the volume and surface area of the 3-D(dimension)figures.

Example: Cube, Cuboid, Cylinder, Cone, Sphere, Frustum, Pyramid, Prism, and Tetrahedron

Volume: The amount of space occupied by the object.

Surface area: The surface area of a solid object is measured by the total area that the surface of the object occupies.

Cube



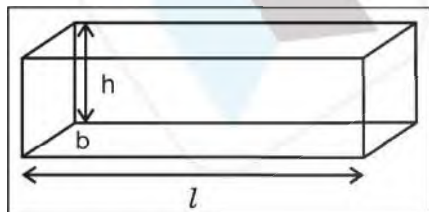
Important Points

- Side = a
- Volume = a^3
- Lateral Surface area = $4a^2$
- Total Surface Area = $6a^2$
- Diagonal = $\sqrt{3}a$

Additional Information

- All sides are equal.
- Cube has all six faces in a square shape.
- Each of the faces meets the other four faces.
- The edges opposite to each other are parallel.

Cuboid



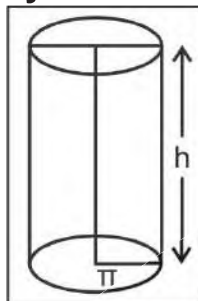
Important Points

- Height = h , Length = l , and Breadth = b
- Volume = $l \times b \times h$
- Volume = Area of base $\times$ height
- Surface area = $2(lb + bh + hl)$
- Diagonal = $\sqrt{l^2 + b^2 + h^2}$
- Area of four walls = Perimeter of base $\times$ height = $2(l + b) \times h$

Additional Information

- Three-dimensional shape with length, breadth, and height
- Each of the faces meets the other four faces.
- The edges opposite to each other are parallel.
- Volume of metal = External volume – internal volume

Cylinder



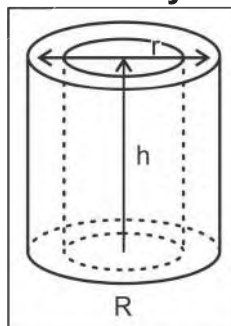
Important Points

- Volume = $\pi r^2 h$
- Surface area = $2\pi r h$
- Total surface area = $2\pi r(h + r)$

Additional Information

- A cylinder has one curved side.
- A cylinder has two vertical flat ends in a circular shape.

Hollow Cylinder



- Curved Surface Area/Lateral Surface Area = $2\pi r h + 2\pi R h = 2\pi h(R + r)$
- Total Surface Area = $2\pi h(R + r) + 2\pi(R^2 - r^2)$
- Volume of hollow cylinder = $\pi r^2 h$
- Volume of material of hollow Cylinder = $\pi(R^2 - r^2)h$

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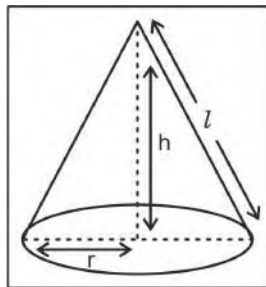
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Cone



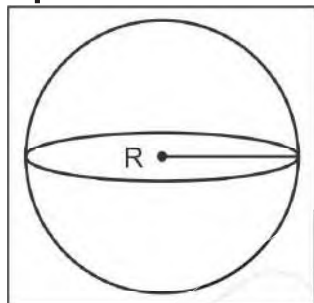
Important Points

- Height = h , Slant height = l , Radius = r
- Volume = $(1/3)\pi r^2 h$
- $l^2 = h^2 + r^2$ Surface area = $\pi r l$
- Total surface area = $\pi r(l + r)$

Additional Information

- It has only one vertex point.
- Its base is circular.

Sphere



For Solid Sphere:

- Volume: $V = \frac{4}{3}\pi r^3$
Where V is the volume and r is the radius of the sphere.
- Surface Area: $A = 4\pi r^2$
Where A is the surface area and r is the radius of the sphere.

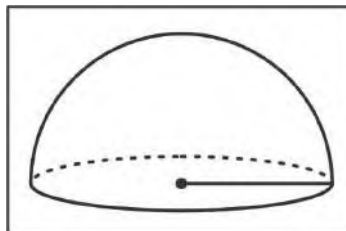
For Hollow Sphere:

- External Surface Area: $A_{\text{ext}} = 4\pi R^2$
Where R is the external radius.
- Internal Surface Area: $A_{\text{int}} = 4\pi r^2$
Where r is the internal radius.
- Total Surface Area of the Hollow sphere:

$$A_{\text{total}} = A_{\text{ext}} + A_{\text{int}}$$

$$A_{\text{total}} = 4\pi R^2 + 4\pi r^2$$

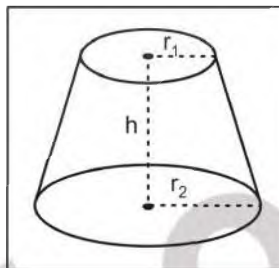
Hemisphere



When r is the radius of the hemisphere

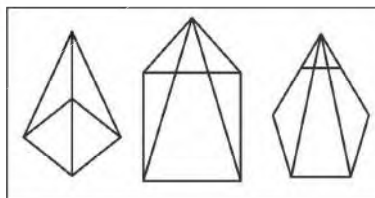
- Volume: $V = \frac{2}{3}\pi r^3$
- Curved Surface Area = $2\pi r^2$
- Total Surface Area = $A_{\text{total}} = 3\pi r^2$

Frustum



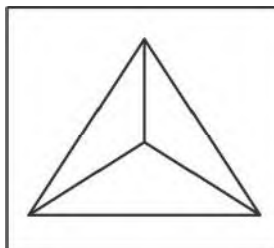
Slant Height (lateral) of a Frustum: $l^2 = h^2 + (r_1 - r_2)^2$
where, l is the slant height, h is the height of the frustum r_1 & r_2 are the radii of the smaller and larger ends of the frustum, respectively.

Pyramid



- Volume = Area of base $\times$ height
- Lateral surface area = Perimeter of base $\times$ slant height
- Total surface area = Perimeter of base $\times$ slant height $\div 2 + 2 \times$ Area of base

Tetrahedron



- Volume = $(a^3\sqrt{2})/12$
- Lateral surface area = $(3\sqrt{3})/4 \times a^2$
- Total surface area = $\sqrt{3}(a^2)$





ALGEBRA

Square Formula:

1. $(a + b)^2 = a^2 + b^2 + 2ab$ or $(a - b)^2 + 4ab$
2. $(a - b)^2 = a^2 + b^2 - 2ab$ or $(a + b)^2 - 4ab$
3. $a^2 + b^2 = (a + b)^2 - 2ab$
4. $a^2 + b^2 = (a - b)^2 + 2ab$
5. $a^2 - b^2 = (a + b)(a - b)$
6. $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
7. $(a + b)^2 - (a - b)^2 = 4ab$
8. $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$
9. $(a^2 - ab + b^2)(a^2 + ab + b^2) = a^4 + a^2b^2 + b^4$
10. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
11. $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca)$
12. $2(ab + bc + ca) = (a + b + c)^2 - (a^2 + b^2 + c^2)$
13. $(b + c)(c + a)(a + b) + abc = (a + b + c)(ab + bc + ca)$
14. $a^2(b - c) + b^2(c - a) + c^2(a - b) = -(b - c)(c - a)(a - b)$
15. $a(b^2 - c^2) + b(c^2 - a^2) + c(a^2 - b^2) = (b - c)(c - a)(a - b)$
16. $a^3(b - c) + b^3(c - a) + c^3(a - b) = -(b - c)(c - a)(a - b)(a + b + c)$

Cube Formula:

1. $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$
2. $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
3. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
4. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$
5. $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$
6. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

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$$7. \quad a^4 - b^4 = (a^2)^2 - (b^2)^2 = (a^2 + b^2)(a^2 - b^2) = (a^2 + b^2)(a + b)(a - b)$$

Additional Information

1: If $a^2 - ab + b^2 = 0$ then $a^3 + b^3 = 0$

2: If $a^2 + a + 1 = 0$ then $a^3 - 1 = 0$ or $a^3 = 1$

3: If $\frac{a}{b} + \frac{b}{a} = 1$ then $a^3 + b^3 = 0$

4: If $\frac{1}{a} - \frac{1}{b} = \frac{1}{a-b}$ then $a^3 + b^3 = 0$

5: If $\frac{a}{b} + \frac{b}{a} = -1$ then $a^3 - b^3 = 0$

6: If $\frac{a}{b} + \frac{b}{a} = \frac{1}{a+b}$ then $a^3 - b^3 = 0$

7: If $ab(a + b) = 1$ then $\frac{1}{a^3 b^3} - a^3 - b^3 = 3$

Factor Formula:

1. $(x + a)(x + b) = x^2 + (a + b)x + ab$

2. $(x - a)(x - b) = x^2 - (a + b)x + ab$

3. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Additional Information

1: If $a = b = c$

then $a^2 + b^2 + c^2 - ab - bc - ca = 0$

thus $a^3 + b^3 + c^3 - 3abc = 0$

Remember, If $a = b = c$ then $a^3 + b^3 + c^3 = 3abc$

2: If $a + b + c = 0$ then

$a^3 + b^3 + c^3 - 3abc = 0$

Remember, If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$



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Function and inverse function:

Square formula:

$$\text{If } x + \frac{1}{x} = a \text{ \& } x - \frac{1}{x} = b$$

$$\Rightarrow x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = \left(x - \frac{1}{x}\right)^2 + 2$$

Remember

$$\Rightarrow x^2 + \frac{1}{x^2} = a^2 - 2 = b^2 + 2$$

$$\Rightarrow x^4 + \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)^2 - 2$$

$$\Rightarrow x^8 + \frac{1}{x^8} = \left(x^4 + \frac{1}{x^4}\right)^2 - 2$$

$$\Rightarrow x + \frac{1}{x} = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 - 2$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = \left(x - \frac{1}{x}\right)^2 + 4$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = \left(x + \frac{1}{x}\right)^2 - 4$$

Remember

$$\Rightarrow x + \frac{1}{x} = \pm\sqrt{b^2 + 4}$$

$$\Rightarrow x - \frac{1}{x} = \pm\sqrt{a^2 - 4}$$

First equation may be given by following type,

$$\Rightarrow x + \frac{1}{x} = 3 \text{ or } x^2 - 3x + 1 = 0$$

$$\text{or } \frac{x^2+1}{x} = 3$$

$$\mathbf{1:} \text{ If } x + \frac{1}{x} = \sqrt{2} \text{ then } x^2 + \frac{1}{x^2} = 0 \text{ or } x^4 + 1 = 0 \text{ or } x^4 = -1$$

$$\text{Remember } \Rightarrow \text{If } x + \frac{1}{x} = \sqrt{2} \text{ then } x^4 + 1 = 0$$

$$\mathbf{2:} \text{ } x + \frac{1}{x} = \pm\sqrt{b^2 + 4} \text{ \& } x - \frac{1}{x} = \pm\sqrt{a^2 - 4}$$

$$x^n + \frac{1}{x^n} = a \text{ then } x^n - \frac{1}{x^n} = \pm\sqrt{a^2 - 4}$$

$$\text{If } x^n - \frac{1}{x^n} = b \text{ then } x^n + \frac{1}{x^n} = \pm\sqrt{b^2 + 4}$$

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Cube Formula:

$$\text{If } x + \frac{1}{x} = a \text{ \& } x - \frac{1}{x} = b$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right)^3 + 3\left(x - \frac{1}{x}\right)$$

Remember

$$\Rightarrow x^3 + \frac{1}{x^3} = a^3 - 3a$$

$$\Rightarrow x^3 - \frac{1}{x^3} = b^3 - 3b$$

$$1: \text{ If } x + \frac{1}{x} = \sqrt{3} \text{ then}$$

$$x^3 + \frac{1}{x^3} = (\sqrt{3})^3 - 3\sqrt{3} = 3\sqrt{3} - 3\sqrt{3} = 0$$

Remember

$$\Rightarrow \text{If } x + \frac{1}{x} = \sqrt{3} \text{ then } x^3 + \frac{1}{x^3} = 0 \text{ or } x^6 = -1$$

$$\text{If } x + \frac{1}{x} = 2 \text{ then } x = 1,$$

$$\text{If } x + \frac{1}{x} = -2 \text{ then } x = -1$$

$$x + \frac{1}{x} = a \Rightarrow x^5 + \frac{1}{x^5} = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 + \frac{1}{x^3}\right) - \left(x + \frac{1}{x}\right)$$

$$x - \frac{1}{x} = a \Rightarrow x^5 - \frac{1}{x^5} = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 - \frac{1}{x^3}\right) - \left(x - \frac{1}{x}\right)$$

$$x + \frac{1}{x} = a \Rightarrow x^6 + \frac{1}{x^6} = \left(x^3 + \frac{1}{x^3}\right)^2 - 2$$

$$x + \frac{1}{x} = a \Rightarrow x^7 + \frac{1}{x^7} = \left(x^4 + \frac{1}{x^4}\right)\left(x^3 + \frac{1}{x^3}\right) - \left(x + \frac{1}{x}\right)$$

$$x + \frac{1}{x} = a \Rightarrow x^7 - \frac{1}{x^7} = \left(x^4 + \frac{1}{x^4}\right)\left(x^3 - \frac{1}{x^3}\right) + \left(x - \frac{1}{x}\right)$$

$$x + \frac{1}{x} = k \Rightarrow x^2 - \frac{1}{x^2} = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right) = \pm k\sqrt{k^2 - 4}$$

$$x - \frac{1}{x} = k \Rightarrow x^2 - \frac{1}{x^2} = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right) = \pm k\sqrt{k^2 + 4}$$

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Componendo & Dividendo:

If $\frac{a}{b} = \frac{x}{y}$ then

$$\frac{a+b}{a-b} = \frac{x+y}{x-y}$$

Special case:

If $x = \frac{2ab}{a+b}$ then

$$\frac{x+2a}{x-2a} + \frac{x+2b}{x-2b} = 2$$

$$\frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}} + \frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}} = 2 \left(\frac{x+y}{x-y} \right)$$

Algebraic Equation:

Algebraic equations are polynomial equations. In examination, generally equations of 1 degree and 2 degree are asked.

Linear Equation:

Polynomial equations with degree 1 i.e. $ax + c = 0$ are called as linear equations.

Ex: linear equations are as follows:

$$2x + 3y = 4$$

$$x + y + z = 10$$

Q. In this question two equations numbered I and II are given. You have to solve both the equations and find out the relation between x and y.

$$\text{I. } 2x + 3y = 13$$

$$\text{II. } 3x + 2y = 12 = 0$$

$$\text{Sol. I. } 2x + 3y = 13 \quad (1)$$

$$\text{II. } 3x + 2y = 12 \quad (2)$$

$(3 \times \text{Equation 2}) - (2 \times \text{Equation 1})$ gives us,

$$\Rightarrow 5x = 10$$

$$\Rightarrow x = 2$$

Putting value of x in equation 1, we get $y = 3$

Hence, $x < y$.

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Q. In the given question, two equations numbered I and II are given. You have to solve both the equations and mark the appropriate answer-

I. $4x + 5y = 14$

II. $2x + 3y = 5$

Sol. $4x + 5y = 14$ (1)

$2x + 3y = 5$ (2)

On multiplying equation (2) by 2, we get,

$4x + 6y = 10$ (3)

Subtracting equation (1) from equation (3),

$y = -4$

$x = 1$ (on putting value of y in the above equation)

$\therefore x > y$.

Quadratic Equation:

Polynomial equations with degree 2 i.e. $ax^2 + bx + c = 0$ are called as quadratic equations.

Some examples of quadratic equations are as follows:

$x^2 + 2x + 3 = 0$

$y^2 - 3y + 4 = 0$

Methods to solve quadratic equation:

Factorisation method

In it quadratic equation $ax^2 + bx + c = 0$ is factorised as $(x - \alpha)(x - \beta) = 0$ and then equation is solved to get $x = \alpha$ or $x = \beta$.

Q. Solve quadratic equation $x^2 - 2x - 15 = 0$

Sol. $x^2 - 2x - 15 = 0$

$\Rightarrow x^2 - 5x + 3x - 15 = 0$

$\Rightarrow x(x - 5) + 3(x - 5) = 0$

$\Rightarrow (x + 3)(x - 5) = 0$

$\Rightarrow x + 3 = 0$ or $x - 5 = 0$

$\Rightarrow x = -3$ or $x = 5$



Sridharachrya's method:

In it quadratic equation $ax^2 + bx + c = 0$ is solved by using formula

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Which gives us,

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ OR } \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Q. Solve quadratic equation $x^2 - 2x - 15 = 0$

Sol. By using Sridharachrya's formula, we get

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \frac{-(-2) + \sqrt{(-2)^2 - 4 \times 1 \times (-15)}}{2 \times 1} = \frac{2 + \sqrt{4 + 60}}{2} = \frac{2 + 8}{2} = 5$$

$$x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \frac{-(-2) - \sqrt{(-2)^2 - 4 \times 1 \times (-15)}}{2 \times 1} = \frac{2 - \sqrt{4 + 60}}{2} = \frac{2 - 8}{2} = -3$$

Q. In the following question two equations are given. You have to solve both the equations and find the relation between x and y.

I. $x^2 = 625$

II. $y = \sqrt{625}$

Sol. We will solve both the equations separately. $x^2 = 625$

$$\Rightarrow x = +25 \text{ or } -25 \text{ (we will consider two values of } x \text{ because of } x^2 \text{) \& } y = \sqrt{625}$$

$$\Rightarrow y = 25 \text{ (The square root is used to refer to only the positive square root) i.e. } [\sqrt{x^2} = |x|]$$

$$\therefore x \leq y$$

Q. In the given question, two equations numbered I and II are given. You have to solve both the equations and find the relation between m and n.

I) $m = \sqrt{324}$

II) $n^2 - 16n - 36 = 0$

Sol. $m = \sqrt{324}$

$$\Rightarrow m = 18$$

$$n^2 - 16n - 36 = 0$$

$$\Rightarrow n^2 - 18n + 2n - 36 = 0$$

$$\Rightarrow n(n - 18) + 2(n - 18) = 0$$

$$\Rightarrow (n - 18)(n + 2) = 0$$

$$\Rightarrow n = (18, -2)$$



Sign Method for Quadratic equations

Let's understand how the sign method plays an important role in finding the sign (+ or -) of roots of any quadratic equation. So, before moving to questions let's go through the below table. "The given table can be used to find the sign of roots for any quadratic equations."

Sr. No	Sign of Equation	Sign of Root (Bigger, Smaller root)
1	+, +	-, -
2	-, +	+, +
3	+, -	-, +
4	-, -	+, -

Q. Find the roots of equation $x^2 + 3x + 2 = 0$

Sol. $x^2 + 3x + 2 = 0$

$$\Rightarrow x^2 + 2x + x + 2 = 0$$

$$\Rightarrow (x + 2)(x + 1) = 0$$

$$\Rightarrow x = -2, -1$$

The above solution is the basic approach to solve quadratic equations.

Q. Find the roots of equation $x^2 - 31x + 240 = 0$

Sol. $x^2 - 31x + 240 = 0$

$$\Rightarrow x^2 - 16x - 15x + 240 = 0$$

$$\Rightarrow (x - 15)(x - 16) = 0$$

$$\Rightarrow x = 15, 16$$

The above solution is the basic approach to solve quadratic equations.

Important Points

Rule #1:

If one equation is $Ax^2 - Bx + C = 0$ while the other equation is $Py^2 + Qy + R = 0$;

then the result will always be $x > y$.

Ex:

I. $x^2 - 18x + 45 = 0$

II. $y^2 + 5y + 6 = 0$



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Equation (1) is of $Ax^2 - Bx + C = 0$ so it will have both the roots positive while equation (2) is of $Py^2 + Qy + R = 0$ it will have both the roots negative. Hence, we can conclude the result will always be $x > y$.

Rule #2:

If the sign of constant term of both the quadratic equations is negative (-ve);
then the result will always be "cannot be determined".

Ex:

I. $x^2 - 5x - 24 = 0$

II. $y^2 - 3y - 40 = 0$

"Since, the sign of constant term is negative in both equations, the answer will be "cannot be determined"

Rule #3:

If both the equations are in the form of squares i.e. $x^2 = 784$ & $y^2 = 900$;
then the result will always be "cannot be determined".



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GEOMETRY

Geometry is a branch of mathematics that deals with the study of shapes, sizes, dimensions, and relationships between different shapes.

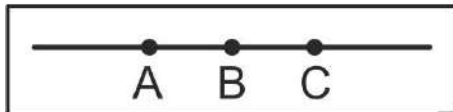
Point:

A point is a specific location in space, like a dot on a map. It doesn't have any size or shape, it's just a single location.

Types of Point:

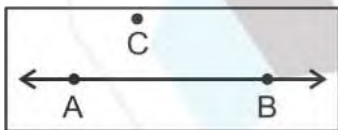
(a) Collinear Points: Collinear points lie on the same line. No matter how many points are on the line, if you can draw a single straight line through all of them, they are considered collinear. For example, if you have points A, B, and C, and you can draw a line that passes through all three of them, then A, B, and C are collinear points.

Ex; Points A, B, and C are collinear.



(b) Non-Collinear Points: Non-collinear points do not lie on the same line. In other words, you cannot draw a single straight line that passes through all of them. For example, if we have three points A, B, and C, and no single line can pass through all three points, then A, B, and C are non-collinear points.

Ex; Points A, B, and C are non-collinear.



Line:

A line is a straight, one-dimensional object that extends infinitely in both directions. It is made up of an infinite number of points that are arranged in a straight path.



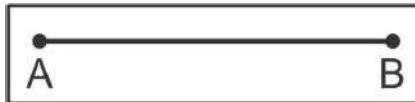
(a) **Straight line:** A line that does not change its direction at any point is called a straight line.

(b) **Curved line:** A line that changes its direction is called a curved line.

Line Segment:

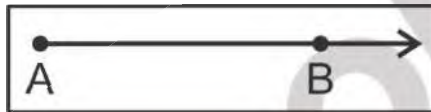
A line segment is part of a line that consists of two end points and all the points between them.

The difference between a line and a line segment is that a line extends infinitely in both directions, while a line segment is a finite portion of a line that has two distinct endpoints.



Ray:

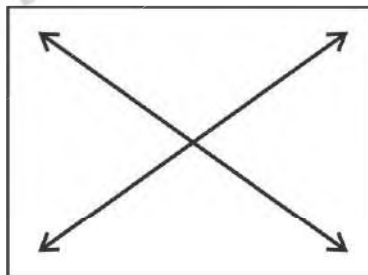
It is part of a line, with one fixed endpoint (also known as the origin) but extending endlessly on the other side.



Types of Lines:

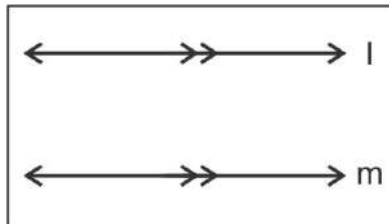
Intersecting Lines:

Intersecting lines are two or more lines that cross each other at a common point, called the point of intersection. The point of intersection is the only point that is shared by both lines, and it is located at the same position on each line.



Parallel Lines:

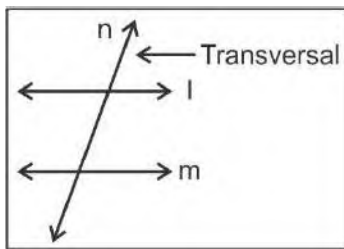
Parallel lines are always equidistant and remain the same distance apart at every point.



Transversal Lines:

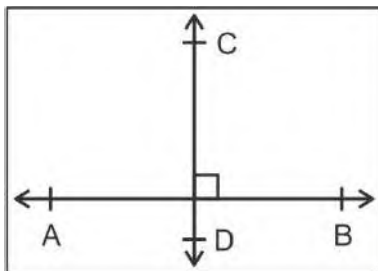
A transversal line is a line that intersects two or more other lines in a plane at different points.





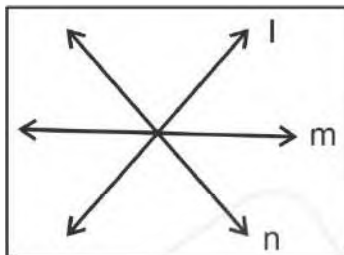
Perpendicular Lines:

Perpendicular lines are two lines that intersect at a right angle (90 degrees). They form four right angles where they intersect.



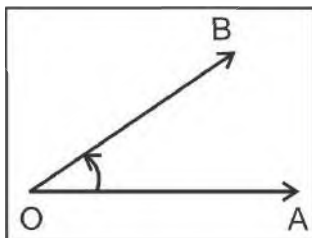
Concurrent Lines:

Concurrent lines are three or more lines that intersect at a common point, called the point of concurrency.



Angle

An angle is made up of two rays, which aren't in a straight line, joined at the same starting point. This starting point is known as the vertex, and the rays are called the arms of the angle. Angles are typically measured in degrees or radians, and they are denoted using the symbol $\angle$.



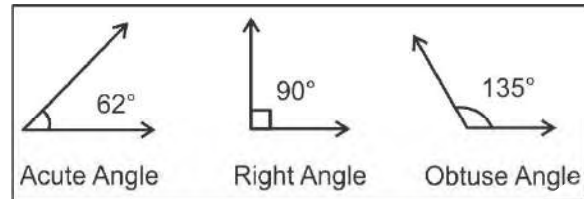
Types of Angles:

Acute Angle: An angle that measures less than 90 degrees (but greater than 0 degrees).

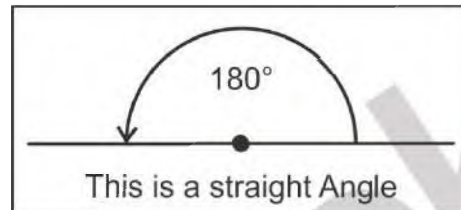
Right Angle: A right angle measures exactly

90 degrees.

Obtuse Angle: An angle that measures greater than 90 degrees (but less than 180 degrees)



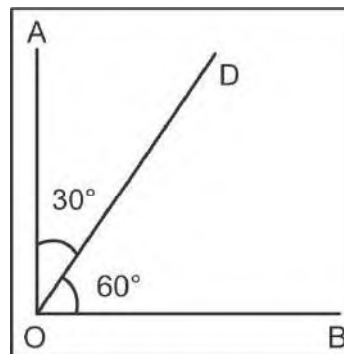
Straight Angle: An angle that measures exactly 180 degrees. It forms a straight line.



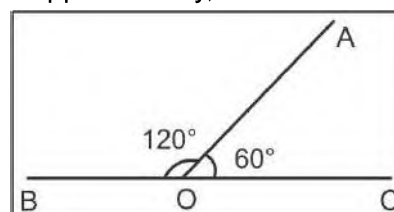
Reflex Angle: An angle that measures greater than 180 degrees (but less than 360 degrees). It extends beyond a straight angle.



Complementary Angles: If angle A and angle B are Complementary, then $A + B = 90$ degrees.



Supplementary Angles: If angle A and angle B are Supplementary, then $A + B = 180$ degrees.



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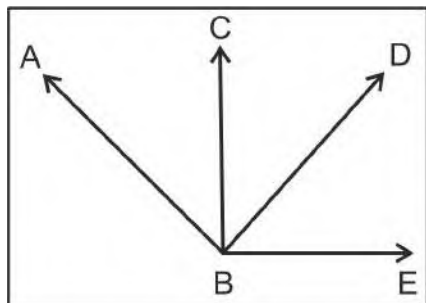


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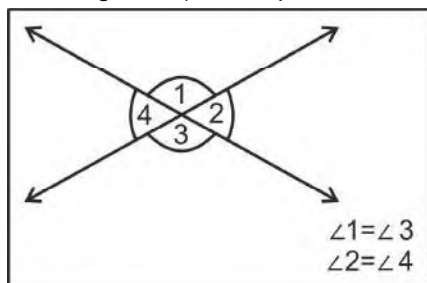
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Adjacent Angles: Adjacent angles are two angles that share a common vertex and a common side, but do not overlap.



Vertical Angles: Vertical angles are a pair of opposite angles formed by the intersection of two lines. They are congruent (have equal measures).



When a transversal line intersects two parallel lines, it creates a series of angles with different properties.

Corresponding Angles 	Examples: $\angle 1$ and $\angle 5$ $\angle 2$ and $\angle 6$ $\angle 3$ and $\angle 7$ $\angle 4$ and $\angle 8$	Alternate Interior Angles 	Examples : $\angle 3$ and $\angle 6$ $\angle 4$ and $\angle 5$
Consecutive Interior Angles 	Examples : $\angle 3$ and $\angle 5$ $\angle 4$ and $\angle 6$	Alternate Exterior Angles 	Examples: $\angle 1$ and $\angle 8$ $\angle 2$ and $\angle 7$
Vertical Angles 	Examples: $\angle 1$ and $\angle 3$ $\angle 2$ and $\angle 4$ $\angle 5$ and $\angle 7$ $\angle 6$ and $\angle 8$	Linear Pair of Angles 	Examples: $\angle 1$ and $\angle 2$ $\angle 2$ and $\angle 3$ $\angle 3$ and $\angle 4$ $\angle 4$ and $\angle 1$ $\angle 5$ and $\angle 6$ $\angle 6$ and $\angle 7$ $\angle 7$ and $\angle 8$ $\angle 8$ and $\angle 5$

- **Alternate Interior Angles:** These are the pair of angles formed inside (between) the parallel lines and on alternate (opposite) sides of the transversal. These angles are equal when the lines are parallel.

- **Alternate Exterior Angles:** These are the pair of angles formed outside (one angle from each of the two lines) of the parallel lines and on alternate (opposite) sides of the transversal, meaning one angle from each parallel line. These angles are also equal when the lines are parallel.

- **Corresponding Angles:** These are the angles that are in the same position on the parallel lines in relation to the transversal. For example, if the transversal intersects the parallel lines at points forming angles at the upper right corner (when viewing the lines from left to right), these angles are corresponding. In parallel lines, corresponding angles are equal.

- **Consecutive (or Co-Interior) Angles:** These are the pair of angles formed inside the parallel lines and on the same side of the transversal. The sum of consecutive interior angles is **180 degrees** (they are supplementary) when the lines are parallel.

- **Vertical (or Opposite) Angles:** These are the angles that are across from each other when two lines intersect. They are not unique to parallel lines and transversal but are commonly observed in this context. Vertical angles are always equal, regardless of whether the intersecting lines are parallel.

- **Linear Pair of Angles:** A linear pair is a pair of adjacent (next to each other) angles formed when two lines intersect. The shared side of the angles lies on a line, hence the term 'linear', and the sum of the angles is always **180 degrees**, meaning they are supplementary. **One important thing to note is that the angles in a linear pair are always adjacent to each other.** The sum of the length of any two sides of a triangle is greater than the length of the third side. The side opposite to the largest angle of a triangle is the largest side.

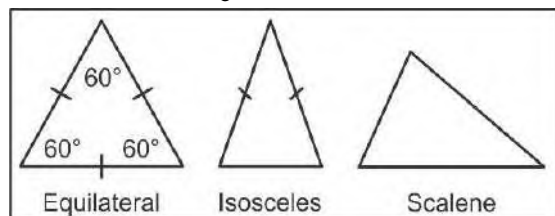


Types of triangles

Triangles can be classified by their sides and their angles:

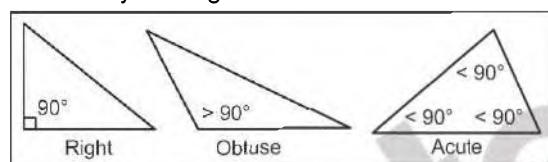
Based on their sides:

1. **Equilateral Triangle:** All three sides of an equilateral triangle are of the same length.
2. **Isosceles Triangle:** An isosceles triangle has two sides of equal length.
3. **Scalene Triangle:** A scalene triangle has all sides of different lengths.

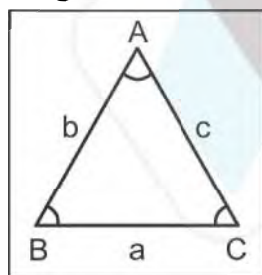


Based on their angles:

1. **Acute Triangle:** All three angles of an acute triangle are less than 90 degrees.
2. **Obtuse Triangle:** One angle of an obtuse triangle is greater than 90 degrees.
3. **Right Triangle:** One angle of a right triangle is exactly 90 degrees.

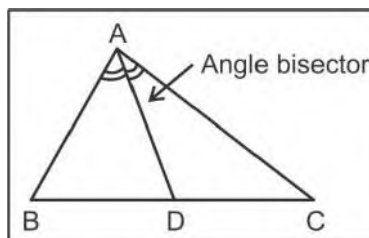


Properties of triangle based on angle.



Sum of all three angles of a triangle is always 180° , i.e. $\angle A + \angle B + \angle C = 180^\circ$

Angle Bisector: A line that divides an angle of a triangle into two equal parts is called an angle bisector of that angle of a triangle.

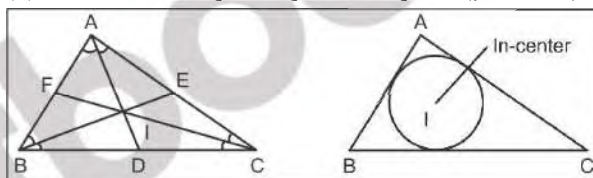


For the angle bisector as shown in the above diagram, $\angle BAD = \angle CAD$

In-center of Triangle: The point of intersection of angle bisectors of a triangle is called the in-centre of a triangle or the centre of the circle which touches the sides of a triangle internally is called in-centre of the triangle $\angle BIC = 90^\circ + \angle A/2$, $\angle AIB = 90^\circ + \angle C/2$, $\angle AIC = 90^\circ + \angle B/2$

Important Points

- (i) Inradius of a triangle $= \Delta/s = \text{Area} / \text{Semi-perimeter}$
- (ii) Inradius of a right-angled triangle $= (p + b - h)/2$



Here, p is perpendicular, b is base and h is hypotenuse

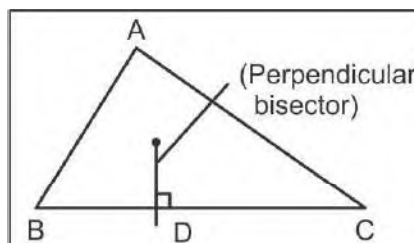
- (iii) Inradius of an equilateral triangle $= \text{Side}/(2\sqrt{3})$
- (iv) Area of triangle

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

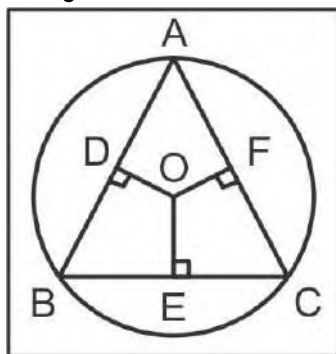
where,

$$s = (a + b + c)/2$$

Perpendicular Bisector: A line that bisects the side of a triangle at an angle of 90° is called a perpendicular bisector.

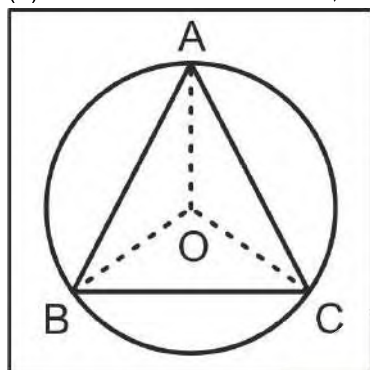


Circumcentre: The circumcenter of a triangle is the point where the perpendicular bisectors of the triangle's sides intersect.



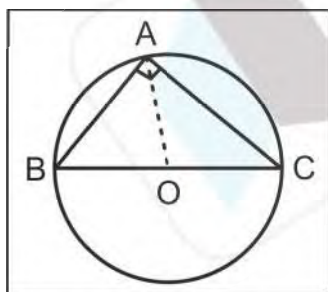
★ Important Points

(a) If O be the circumcentre, then



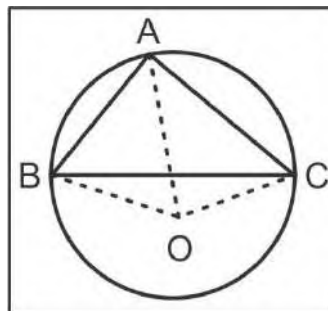
In an acute-angled triangle ABC, $\angle BOC = 2\angle A$, $\angle COA = 2\angle B$ & $\angle AOB = 2\angle C$

In a right-angled triangle, ABC



$$\begin{aligned}\angle BOC &= 2\angle A = 180^\circ \\ \angle COA &= 2\angle B = 180^\circ - 2\angle C \\ \angle AOB &= 2\angle C = 180^\circ - 2\angle B\end{aligned}$$

In an obtuse-angled triangle ABC



$$\begin{aligned}\angle BOC &= 2(180^\circ - \angle A) \\ \angle COA &= 2\angle B \text{ \& } \angle BOA = 2\angle C\end{aligned}$$

★ Important Points

In a acute triangle, the circumcentre is **always** inside the triangle.

In a right-angled triangle, the circumcentre is mid-point of the hypotenuse

In an obtuse-angled triangle, the circumcentre is a ways outside the triangle and **will** be at the front of the angle which is obtuse.

The circumradius of any triangle ABC =

Multiplication of **all** three sides /

$$(4 \times \text{Area of the triangle}) = abc/4\Delta$$

The circumradius of a right-angled triangle

$$= \text{Hypotenuse}/2 = h/2$$

The circumradius of an equilateral triangle = Side/ $\sqrt{3}$

$$\text{Note:- } 2(r + R) = a + b$$

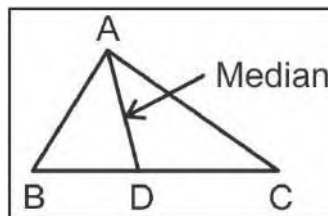
Where a and b are the base and perpendicular of right angle triangle.

Distance between the circum-centre and incentre in any triangle is $(R^2 - 2Rr)$

Where R = circum-radius r = inradius

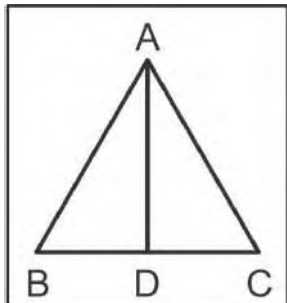
Excentre: The intersection point of internal angle bisector of one angle and bisectors of other two opposite exterior angles.

Median of a triangle: A line joining the vertices of a triangle to the middle point of the opposite side of a triangle is called the median of the triangle.



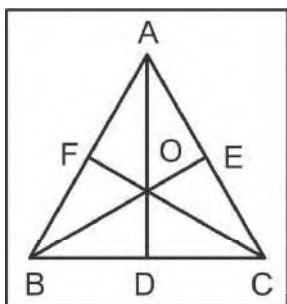
Centroid of a triangle: The point of intersection of the medians of a triangle is called the centroid of a triangle. If G is the centroid of a triangle and D, E, and F are the midpoints of BC, AC, and AB, then $AG:GD = BG:GE = CG:GF = 2:1$

(a) Apollonius Theorem



$$AB^2 + AC^2 = 2(AD^2 + CD^2)$$

The sum of any two sides of a triangle is greater than twice the median drawn to the third side.



$$(AB + AC > 2AD \dots (1) \quad AB + BC > 2BE \dots (2) \quad AC + BC > 2CF \dots (3))$$

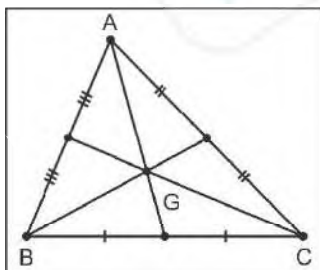
$$\text{Adding (1), (2) and (3)}$$

$$2(AB + AC + BC) > 2(AD + BE + CF)$$

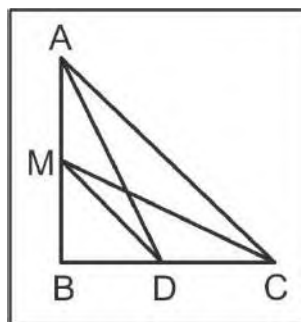
$$AB + AC + BC > AD + BE + CF$$

Sum of sides (perimeter) is always greater than the sum of all median.

$$3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)$$



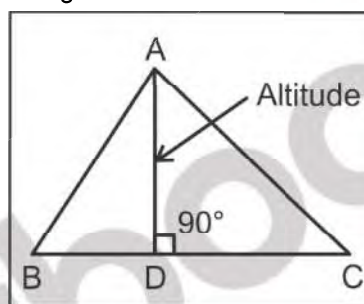
$$AB^2 + GC^2 = AC^2 + BG^2$$



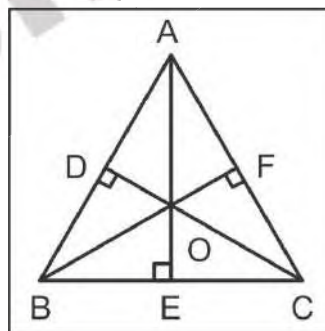
Here, AD and MC are Median of triangle ABC

$$AD^2 + MC^2 = AC^2 + MD^2$$

Altitude of a triangle: Perpendicular lines drawn from the vertices of a triangle to the opposite sides of a triangle are called the altitudes of a triangle.



Orthocenter is the point of intersection of the altitudes of a triangle.



If O is the orthocenter of a triangle, then $\angle BOC + \angle A = 180^\circ$, $\angle AOB + \angle C = 180^\circ$, $\angle AOC + \angle B = 180^\circ$.

★ Important Points

In an acute-angled triangle, the orthocentre is always inside the triangle.

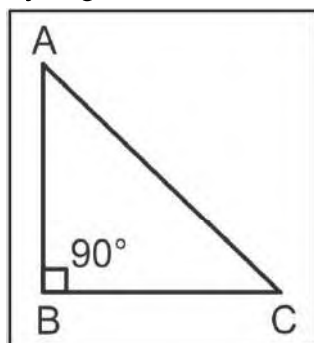
In a right-angled triangle, the orthocentre is on the vertex of the triangle at which the triangle is right-angled.

In an obtuse-angled triangle, the orthocentre is always outside of the triangle and will be at the back of the angle which is obtuse.



Theorems of Triangles:

Pythagoras Theorem:



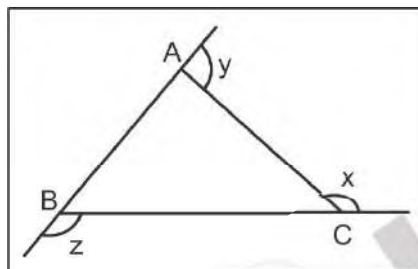
In any right-angled triangle $AB^2 + BC^2 = AC^2$, where $\angle B = 90^\circ$

★ Important Points

Properties of Triangle-based on Angles:

An exterior angle of a triangle is equal to the sum of opposite interior angles of a triangle.

Here,

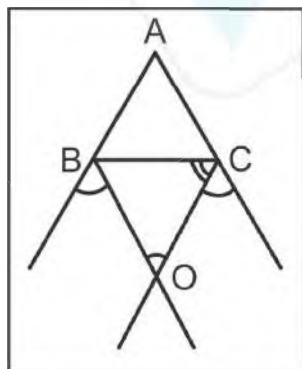


$$\angle X = \angle A + \angle B, \angle Y = \angle B + \angle C, \angle Z = \angle A + \angle C$$

Sum of the exterior angle of a triangle is 360°

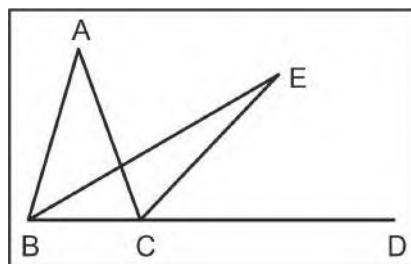
$$\text{Here, } \angle X + \angle Y + \angle Z = 360^\circ$$

If in an $\triangle ABC$, the sides AB and AC are extended, and the bisectors of exterior angles of $\angle B$ and $\angle C$ Meet at O,



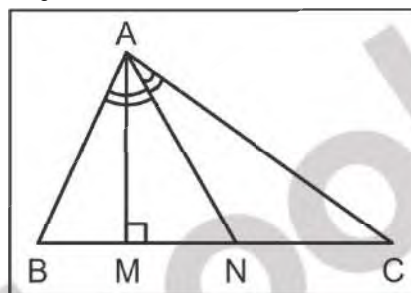
$$\text{then, } \angle BOC = 90^\circ - \angle A/2$$

In an $\triangle ABC$, if BC is extended to D, and BE and CE are the bisectors of $\angle ABC$ and $\angle ACD$ which meet at E,



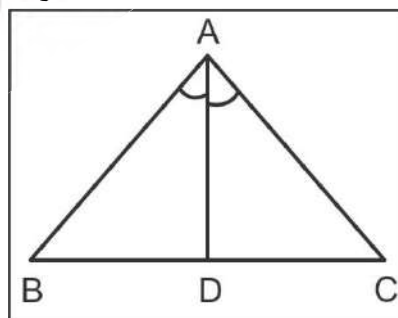
$$\text{then, } \angle BEC = 1/2 \angle BAC$$

In $\triangle ABC$, if AM is an altitude on BC and AN is the angle bisector of $\angle A$



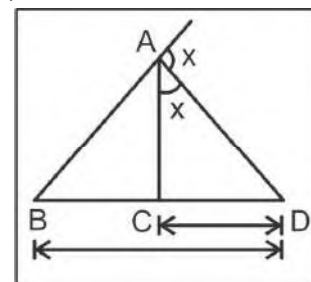
$$\text{then, } \angle MAN = 1/2 (\angle B - \angle C)$$

Internal Angle bisector theorem: The Internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle.



$$BD/CD = AB/AC$$

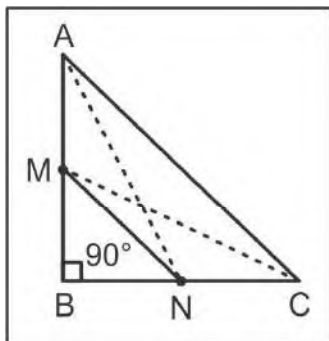
Exterior angle bisector theorem: In $\triangle ABC$, if AD is the angle bisector of exterior $\angle A$ which meets produced BC at D.



$$\text{then, } AB/AC = BD/CD$$

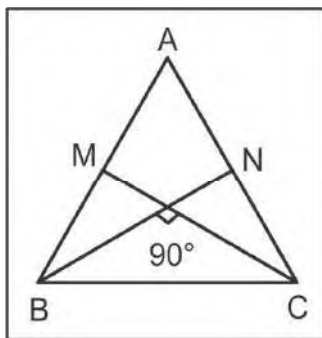


In a right-angled $\triangle ABC$ right angled at B, if the midpoint of AB and BC are M and N respectively,



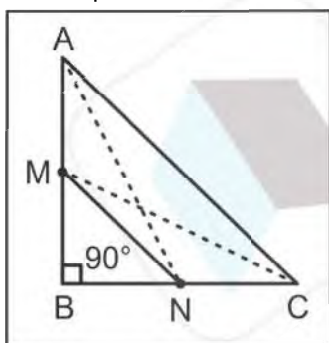
then, $4[AN^2 + CM^2] = 5AC^2 = 20MN^2$ $AC^2 = 4 MN^2$

In $\triangle ABC$, if BN and CM are the medians that intersect at 90° ,



then, $AB^2 + AC^2 = 5 BC^2$

In a right angle, $\triangle ABC$ right angled at B, if M and N are the midpoints of side AB and BC respectively,



then, $AN^2 + CM^2 = AC^2 + MN^2$

Similarity and Congruence in Triangle

Similarity: Similar figures maintain the same shape but not necessarily size.

Congruency: Congruent figures are exact replicas of each other.

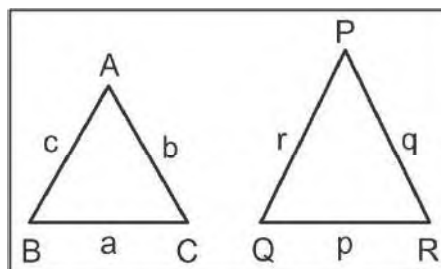
	Similarity($\sim$)	Congruency($\cong$)
Definition	Same shape, different size.	Same shape, same size.
Sides	Proportional.	Equal.
Angles	Equal.	Equal.

Properties for Similarity & Congruency:

Postulate	Similarity	Congruency
AA	Two equal angles, triangles similar.	-
SAS	Proportional sides around the equal angle, triangles similar.	Equal sides around equal angles, triangles congruent.
SSS	Proportional sides, triangles similar.	Equal sides, triangles be congruent.
ASA	-	Two angles and included side equal, triangles congruent.
AAS	-	Two angles and non-included sides equal, triangles congruent.

(A) Properties of Similar triangles

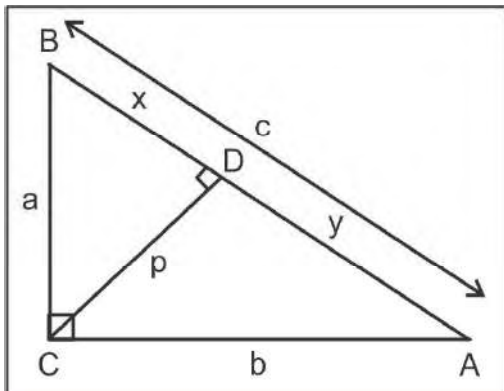
If $\triangle ABC$ and $\triangle PQR$ are similar, then



- (i) $a/p = b/q = c/r$
 (ii) Ratio of corresponding sides = Ratio of perimeter = Ratio of semi-perimeter(s) = Ratio of corresponding medians = Ratio of inradius = Ratio of circumradius
 (iii) Ratio of area = (Ratio of corresponding sides)²

(B) In a right-angled triangle ABC,

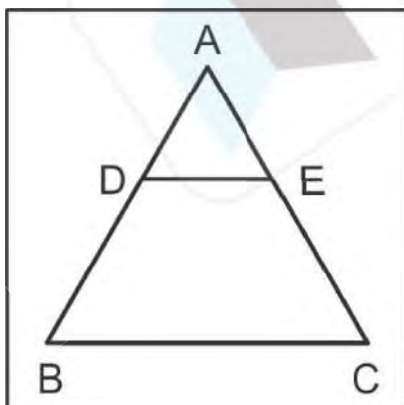
if a perpendicular (CD) is drawn from the vertex which is right angle (C) to the hypotenuse (AB), then



- (i) (a) $a^2 = x \times c$
 (b) $b^2 = y \times c$
 (ii) $p^2 = x \times y$
 (iii) $a \times b = p \times c$
 (iv) $1/p^2 = 1/a^2 + 1/b^2$

Basic Proportionality Theorem (Thales Theorem)

1. A line drawn parallel to one side of a triangle divides the other two sides in the same ratio.
2. If a line divides any two sides of a triangle in the same ratio, the line must be parallel to the third side.

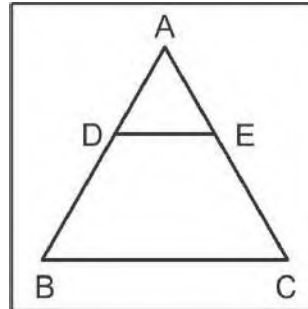


In $\triangle ABC$,

If $DE \parallel BC$, then $AD/DB = AE/EC$

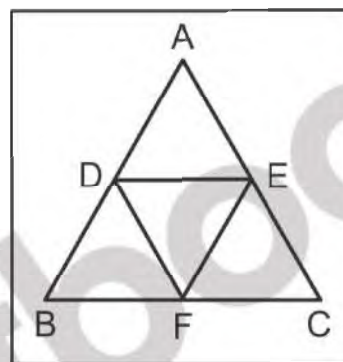
or If $AD/DB = AE/EC$, then $DE \parallel BC$

Mid-point Theorem:



If D and E are the midpoints of side AB and AC, then, $DE = \frac{1}{2} BC$ and $DE \parallel BC$.

★ Important Points



If D, E, and F are the midpoints of the sides AB, AC, and BC of $\triangle ABC$,

then, $DE = \frac{1}{2} BC$, $DF = \frac{1}{2} AC$, $EF = \frac{1}{2} AB$, and
 $\text{ar}(\triangle ADE) = \text{ar}(\triangle BDF) = \text{ar}(\triangle DEF) = \text{ar}(\triangle EFC)$
 $= \frac{1}{4} \text{ar}(\triangle ABC)$

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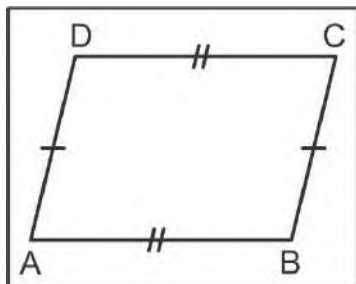
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Quadrilaterals:

The sum of angles of any quadrilateral is 360°

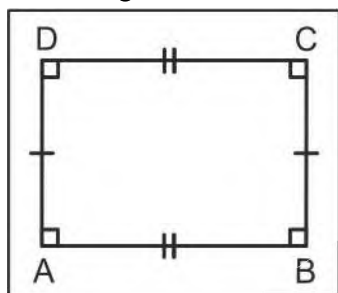
Types of Quadrilateral:

1. Parallelogram



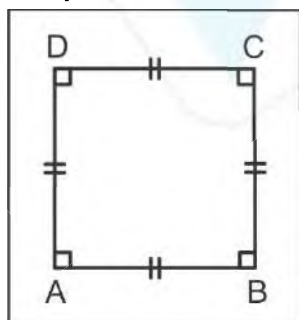
- Opposite sides are equal and parallel.
- Opposite angles are equal.
- Consecutive angles are supplementary.
- Diagonals bisect each other.
- Each diagonal forms two congruent triangles.

2. Rectangle:



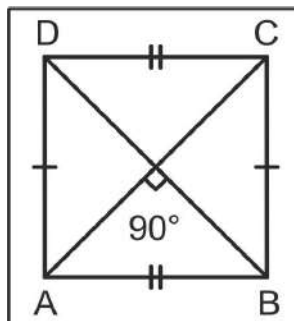
- All sides are parallel, with opposite sides being equal.
- All angles are 90 degrees.
- Diagonals are equal and bisect each other.
- Each diagonal forms two congruent triangles.

3. Square:



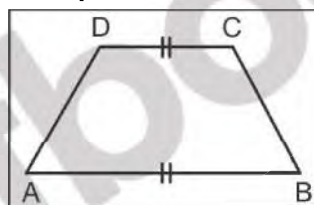
- All sides are equal and parallel.
- All angles are 90 degrees.
- Diagonals are equal, bisect each other, and form 90-degree angles.
- Each diagonal forms two congruent triangles.

4. Rhombus:



- All sides equal in length.
- Opposite angles equal.
- Diagonals bisect each other at right angles.
- Diagonals divide rhombus into 4 congruent triangles.
- Sum of interior angles is 360 degrees.
- Each diagonal divides into two equilateral triangles.
- Opposite sides are parallel.

5. Trapezium



- Has one pair of opposite sides parallel.
- The sum of interior angles is 360 degrees.
- Non-parallel sides are not equal in general.
- Diagonals intersect at different angles, not necessarily at 90 degrees.
- In an isosceles trapezium, diagonals are equal in length, and base angles are equal.

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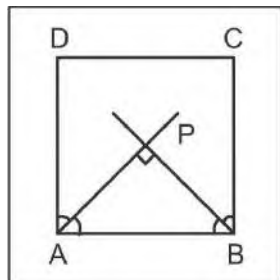
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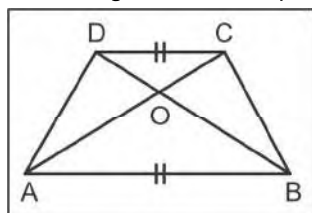
Additional Information

In a quadrilateral ABCD, if the angle bisectors of $\angle A$ and $\angle B$ meet at P,



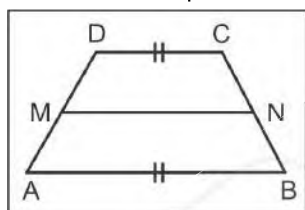
then, $\angle APB = (\angle C + \angle D)/2$

If the diagonals of a trapezium ABCD meet at O,



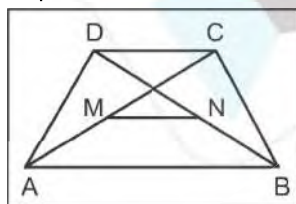
then, $OA \times OC = OB \times OD = AB \times CD$

If M and N are the midpoints of non-parallel sides AD and BC of a trapezium ABCD in which $AB \parallel CD$



then, $MN = \frac{1}{2} (AB + CD)$

If M and N are the midpoints of diagonals of a trapezium ABCD in which $AB \parallel CD$,



then, $MN = \frac{1}{2} (AB - CD)$, where $AB > CD$

Polygons:

- Triangle: 3-sided polygon.
- Quadrilateral: 4-sided polygon.
- Pentagon: 5-sided polygon.
- Hexagon: 6-sided polygon.

Important Points

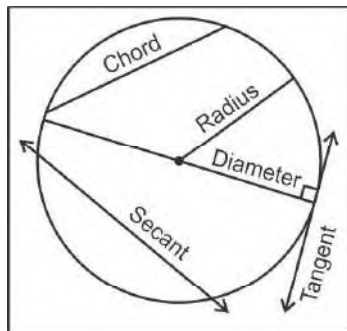
- Sum of interior angles of a polygon** = $(2n - 4) \times 90^\circ$, where n = number of sides
- Each exterior angle of a regular polygon** = $360^\circ/n$, where n = number of sides
- Each interior angle of a regular polygon** = $180^\circ - \text{Exterior angle of polygon}$
- The angle made by side of a polygon at the center of the regular polygon**,
= $360^\circ/n$, where n is a number of sides
- Number of diagonals of a polygon** = $n(n - 3)/2$, where n = number of sides
- The sum of all exterior angle of polygon** = 360°
- Area of a regular polygon** = $(na^2/4) \times \cot(\pi/n)$, where n = number of sides, a = side length
- Radius of the circumcircle of a regular polygon** = $(a/2) \times \text{cosec}(\pi/n)$, where n = Number of sides, a = side length
- Radius of in-centre of a regular polygon** = $(a/2) \times \cot(\pi/n)$, where n = Number of sides, a = side length

Additional Information

Name of Polygon	Number of Sides	Sum of Internal Angles (degrees)	Internal Angle (degrees)	External Angle (degrees)	Number of Diagonals
Regular Triangle	3	180	60	120	0
Regular Quadrilateral	4	360	90	90	2
Regular Pentagon	5	540	108	72	5
Regular Hexagon	6	720	120	60	9
Regular Heptagon	7	900	Approx. 128.57	Approx. 51.43	14

Remember that these internal and external angles apply to regular polygons where all sides and angles are equal.

Circle:



Center: The fixed point in the plane from which all points on the circle are equidistant.

Radius: The distance between the center of the circle and any point on the circle.

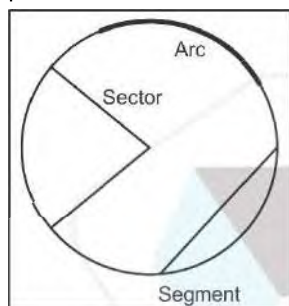
Diameter: The distance across the circle, passing through the center (twice the radius).

Circumference: The boundary or perimeter of the circle.

Tangent: A line that touches the circle at exactly one point.

Chord: A line segment connecting two points on the circle.

Secant: A line that intersects the circle at two distinct points.



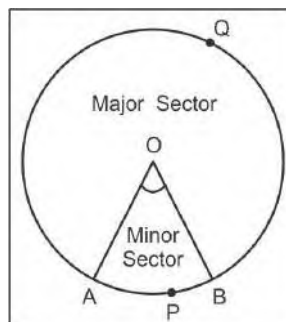
Arc: A portion of the circumference of the circle.

Sector: The region enclosed by two radii and the corresponding arc.

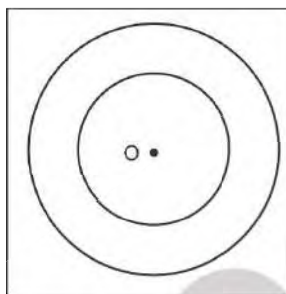
Major Sector: A sector that covers more than half of the circle's circumference.

Minor Sector: A sector that covers less than half of the circle's circumference.

Segment: A region enclosed by a chord and the corresponding arc

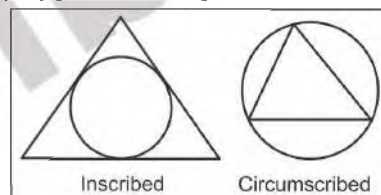


Concentric circles: Circles with the same centre but different radii.



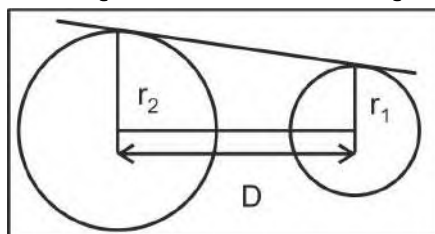
Inscribed circle: A circle inside a polygon touching each side of the polygon.

Circumscribed circle: A circle drawn around a polygon, touching each vertex of the polygon.



★ Important Points

When two circles do not touch each other, then the Length of direct common tangent is,

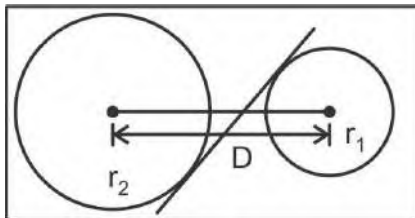


where d = distance between the centre of the circle and r_2 and r_1 are the radii of the circles

$$\sqrt{d^2 - (r_2 - r_1)^2}$$



When two circles do not touch each other, then the length of the transverse common tangent is

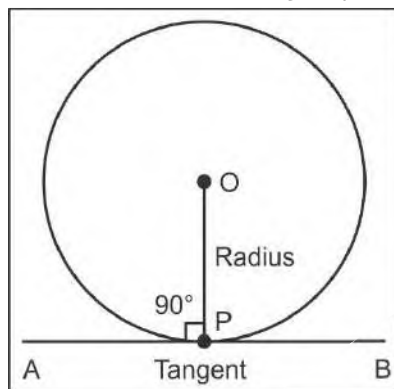


where d = distance between the centre of the circle and r_2 and r_1 are the radii of the circles

$$\sqrt{d^2 - (r_2 + r_1)^2}$$

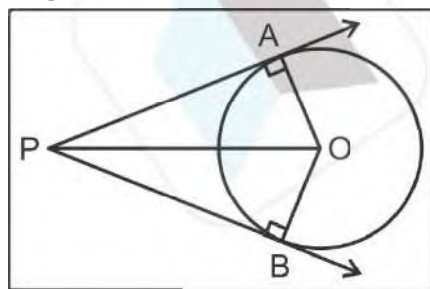
Tangent Radius Theorem:

The tangent makes a right angle with the radius of a circle at the point of tangency.



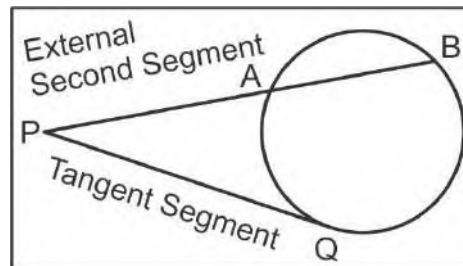
Two Tangents Theorem

The Two Tangent Theorem states that from an external point, two tangents drawn to a circle are equal in length.

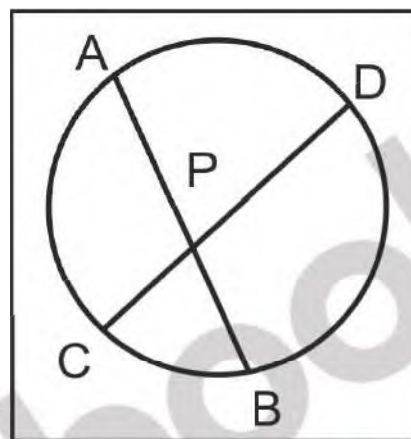


$$PA = PB$$

Tangent-Secant theorem:

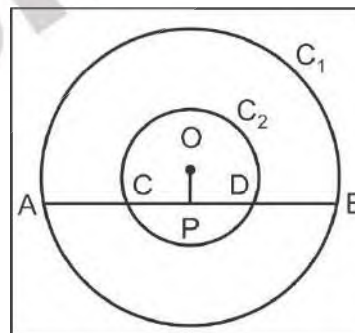


$$PQ^2 = PA \times PB$$



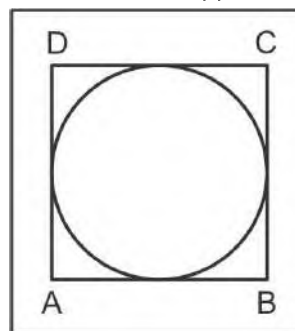
$$PA \times PB = PC \times PD$$

If c_1 and c_2 are two concentric circles and AB is a common chord and point P is such that $OP \perp AB$,



then, $AC = BD$

If a circle touches all four sides of a quadrilateral then the sum of the opposite sides is equal.

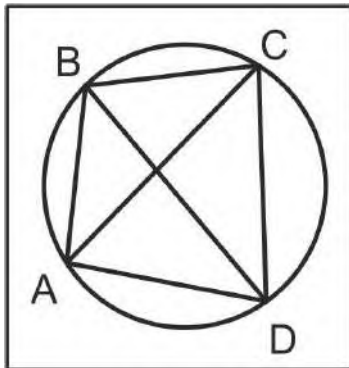


$$AB + DC = AD + BC$$



Cyclic Quadrilateral:

A cyclic quadrilateral is a quadrilateral that can be inscribed within a circle.



- All four vertices of a cyclic quadrilateral lie on a common circle.
- The opposite angles of a cyclic quadrilateral are supplementary (their sum is 180 degrees).
- The exterior angle formed by extending one side of a cyclic quadrilateral is equal to the interior opposite angle.
- The opposite sides of a cyclic quadrilateral are in a ratio of their lengths.

★ Important Points

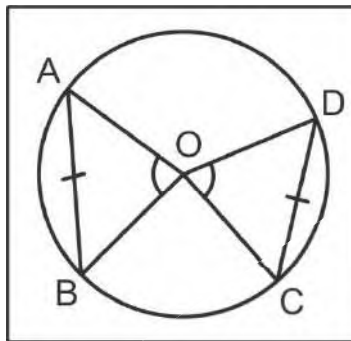
Ptolemy's Theorem: If ABCD is a cyclic quadrilateral, the theorem states that the product of the diagonals' lengths is equal to the sum of the products of opposite sides:

$$AC \times BD = AB \times CD + AD \times BC$$

- The intersection point of the diagonals of a cyclic quadrilateral lies on the circle's circumference.
- The sum of the squares of the lengths of the diagonals of a cyclic quadrilateral is equal to the sum of the squares of its side lengths.
- The perpendicular bisectors of the sides of a cyclic quadrilateral intersect at a single point (the circumcenter).

Properties and Theorems Related to Circles

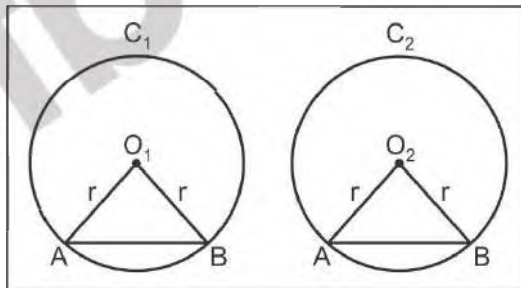
1. If two chords in a circle are of equal length, the subtend equal angles at the centre of the circle, and conversely, if two chords subtend equal angles at the centre, then the chords themselves are of equal length.



If $AB = CD$, then $\angle AOB = \angle COD$

If $\angle AOB = \angle COD$, then $AB = CD$

2. In two equal circles, equal chords subtend equal angles at the centres; similarly, if the angles subtended by the chords at the centres of the equal circles are equal, then the corresponding chords are equal.



If C_1 and C_2 are equal circles and $AB = PQ$, then $\angle AOB = \angle POQ$

If C_1 and C_2 are equal circles and $\angle AOB = \angle POQ$, then $AB = PQ$.

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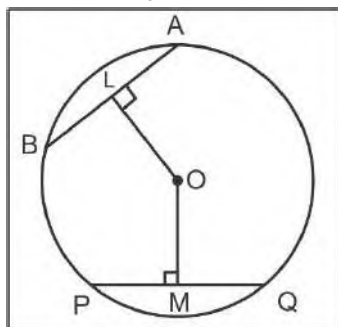
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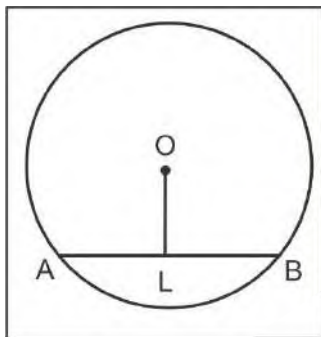
3. Equal chords of a circle are equidistant from the centre, and conversely, if two chords of a circle are equidistant from the centre, then those chords are equal in length.



If $AB = PQ$, then $OL = OM$

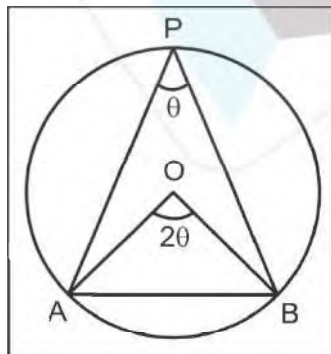
If $OL = OM$, then $AB = PQ$

4. A perpendicular from the centre of a circle to a chord bisects the chord.



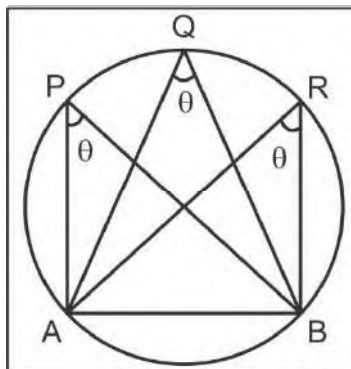
If $OL \perp AB$, then $AL = LB$.

5. The angle formed at the centre of a circle by an arc is twice the angle that the same arc forms at any other point on the circle's circumference that lies on the same side as the centre.



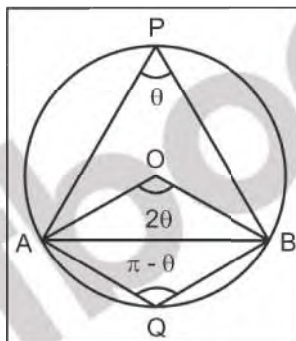
$\angle AOB = 2 \times \angle APB$

6. The angle in the same segment of a circle are equal i.e.



$\angle APB = \angle AQB = \angle ARB$

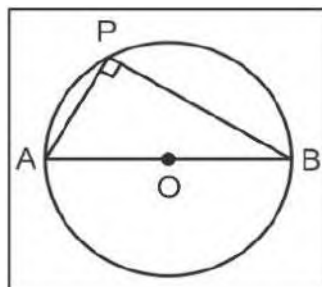
7. If AB is a chord, O is the centre and P and Q are any points in the major and the minor segments of the circle respectively, then



If $\angle APB = \theta$, then $\angle AOB = 2\theta$ and $\angle AQB = \pi - \theta$

The angles in the major segment and the minor segment are supplementary.

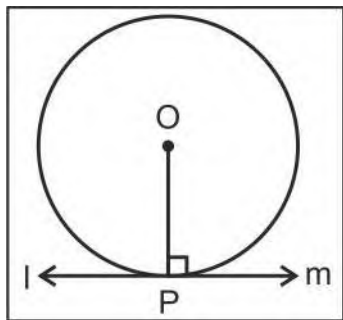
8. The angle of a semicircle is a right angle.



If AB is a diameter and P is any point on the circumference, then $\angle APB = 90^\circ$

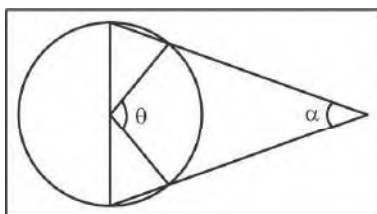


9. The tangent at any point on a circle is perpendicular to the radius drawn through the point of contact.



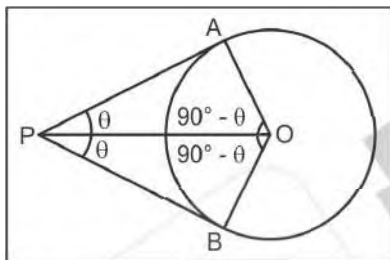
$$\angle OPM = \angle OPI = 90^\circ$$

10.



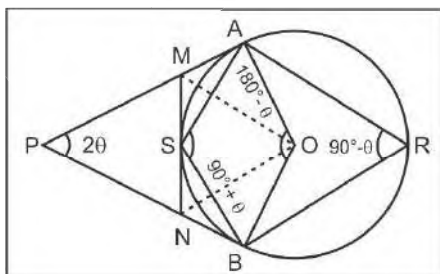
$$\angle \theta = 180 - 2\alpha$$

11. Two tangent PA and PB are drawn from an external point P on a circle, whose centre is O, then



- $\triangle PAO \cong \triangle PBO$
- $PA = PB$
- $\angle APO = \angle BPO$
- $\angle PAO = \angle PBO$

12. PA and PB are two tangents, O is the center of the circle and R and S are the points on the circle, then



Let $\angle APB = 2\theta$, then

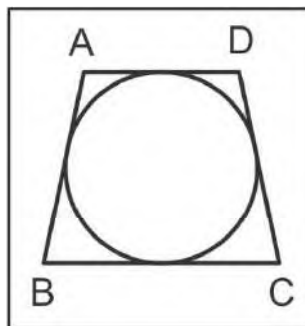
$$\angle AOB = 180^\circ - 2\theta$$

$$\angle ARB = 90^\circ - \theta$$

$$\angle ASB = 90^\circ + \theta$$

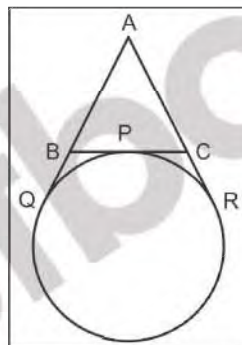
$$\angle AOB = 2\angle MON$$

13. If a quadrilateral ABCD circumscribes a circle,



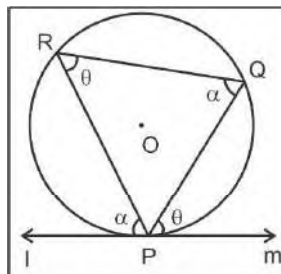
then, $AB + CD = BC + AD$

14. A circle externally touch the side BC of a $\triangle ABC$ at P, AB produced at Q and AC produced at R, then the perimeter of $\triangle ABC$,



If $Q = x$ cm then Perimeter $= 2 \times AQ = 2x$

15. A chord is drawn through the point of contact of a tangent, and then the angles which the chord makes with the tangent are equal to the angle made by that chord in the alternate segment



Let $\angle MPQ = \theta$,

then $\angle PRQ = \theta$ and let $\angle IPR = \alpha$, then $\angle PQR = \alpha$



TRIGONOMETRY

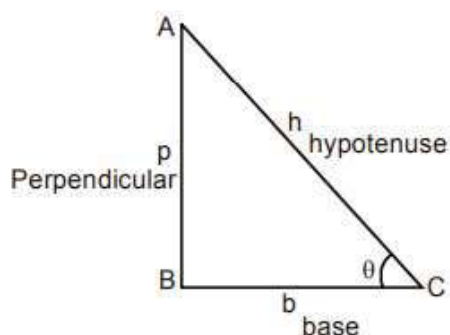
Trigonometry is a branch of mathematics that studies relationships involving the lengths and angle of triangles.

Trigonometric Ratios

There are six trigonometric ratios:

sine (sin), cosine (cos), tangent (tan), cosecant (cosec), secant (sec), and cotangent (cot).

For a right-angled triangle, these are defined as:



$$(I) \sin\theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{P}{H}$$

$$(II) \cos\theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{B}{H}$$

$$(III) \tan\theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{P}{B}$$

$$(IV) \text{cosec}\theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{H}{P}$$

$$(V) \sec\theta = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{H}{B}$$

$$(VI) \cot\theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{B}{P}$$

Relations Between Trigonometric Ratios

$$(I). (a) \sin\theta = \frac{1}{\text{cosec}\theta}$$

$$(b) \text{cosec}\theta = \frac{1}{\sin\theta}$$

$$(c) \sin\theta \cdot \text{cosec}\theta = 1$$

(II).

$$(a) \cos\theta = \frac{1}{\sec\theta}$$

$$(b) \sec\theta = \frac{1}{\cos\theta}$$

$$(c) \cos\theta \cdot \sec\theta = 1$$

(III).

$$(a) \tan\theta = \frac{1}{\cot\theta}$$

$$(b) \cot\theta = \frac{1}{\tan\theta}$$

$$(c) \tan\theta \cdot \cot\theta = 1$$

(IV).

$$(a) \tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$(b) \sin\theta = \tan\theta \cdot \cos\theta$$

$$(c) \cos\theta = \frac{\sin\theta}{\tan\theta}$$

(V).

$$(a) \cot\theta = \frac{\cos\theta}{\sin\theta}$$

$$(b) \cos\theta = \cot\theta \cdot \sin\theta$$

$$(c) \sin\theta = \frac{\cos\theta}{\cot\theta}$$



Trigonometric Ratios of Some Special Angles

Values of the trigonometric ratios for some special angles are given below

	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/√2	0
tan	0	1/√3	1	√3	∞
cosec	∞	2	√2	2/√3	1
sec	1	2/√3	√2	2	∞
cot	∞	√3	1	1/√3	0

Co - Function Identities

For any angle θ ,

$\sin(90^\circ - \theta) = \cos \theta$	$\cos(90^\circ - \theta) = \sin \theta$
$\tan(90^\circ - \theta) = \cot \theta$	$\cot(90^\circ - \theta) = \tan \theta$
$\sec(90^\circ - \theta) = \operatorname{cosec} \theta$	$\operatorname{cosec}(90^\circ - \theta) = \sec \theta$

★ Important Points

- 1) If $\sin \theta + \operatorname{cosec} \theta = 2$, then $\sin^n \theta + \operatorname{cosec}^n \theta = 2$
- 2) If $\cos \theta + \sec \theta = 2$, then $\cos^n \theta + \sec^n \theta = 2$
- 3) If $\tan \theta + \cot \theta = 2$, then $\tan^n \theta + \cot^n \theta = 2$
- 4) If $(\sec \theta + \tan \theta) = x$, then $(\sec \theta - \tan \theta) = 1/x$
- 5) If $(\sec \theta - \tan \theta) = x$, then $(\sec \theta + \tan \theta) = 1/x$
- 6) If $(\operatorname{cosec} \theta + \cot \theta) = x$, then $(\operatorname{cosec} \theta - \cot \theta) = 1/x$
- 7) If $(\operatorname{cosec} \theta - \cot \theta) = x$, then $(\operatorname{cosec} \theta + \cot \theta) = 1/x$

Trigonometric Identities

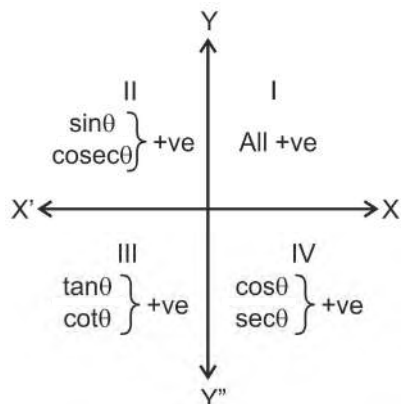
$\sin^2 \theta + \cos^2 \theta = 1$	$\sin 2\theta = 2 \sin \theta \cos \theta$
$\tan^2 \theta + 1 = \sec^2 \theta$	$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$
$\cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$	$\tan 2\theta = 2 \tan \theta / (1 - \tan^2 \theta)$

Additional Information

	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	$\operatorname{cosec} \theta$
$\sin \theta$	$\sin \theta$	$\sqrt{1 - \cos^2 \theta}$	$\frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}}$	$\frac{1}{\sqrt{1 + \cot^2 \theta}}$	$\frac{\sqrt{\sec^2 \theta - 1}}{\sec \theta}$	$\frac{1}{\operatorname{cosec} \theta}$
$\cos \theta$	$\sqrt{1 - \sin^2 \theta}$	$\cos \theta$	$\frac{1}{\sqrt{1 + \tan^2 \theta}}$	$\frac{\cot \theta}{\sqrt{1 + \cot^2 \theta}}$	$\frac{1}{\sec \theta}$	$\frac{\sqrt{\operatorname{cosec}^2 \theta - 1}}{\operatorname{cosec} \theta}$
$\tan \theta$	$\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$	$\frac{\sqrt{1 - \cos^2 \theta}}{\cos \theta}$	$\tan \theta$	$\frac{1}{\cot \theta}$	$\sqrt{\sec^2 \theta - 1}$	$\frac{1}{\sqrt{\operatorname{cosec}^2 \theta - 1}}$
$\cot \theta$	$\frac{\sqrt{1 - \sin^2 \theta}}{\sin \theta}$	$\frac{\cos \theta}{\sqrt{1 - \cos^2 \theta}}$	$\frac{1}{\tan \theta}$	$\cot \theta$	$\frac{1}{\sqrt{\sec^2 \theta - 1}}$	$\sqrt{\operatorname{cosec}^2 \theta - 1}$
$\sec \theta$	$\frac{1}{\sqrt{1 - \sin^2 \theta}}$	$\frac{1}{\cos \theta}$	$\frac{\sqrt{1 + \tan^2 \theta}}{\tan \theta}$	$\frac{\sqrt{1 + \cot^2 \theta}}{\cot \theta}$	$\sec \theta$	$\frac{\operatorname{cosec} \theta}{\sqrt{\operatorname{cosec}^2 \theta - 1}}$
$\operatorname{cosec} \theta$	$\frac{1}{\sin \theta}$	$\frac{1}{\sqrt{1 - \cos^2 \theta}}$	$\frac{\sqrt{1 + \tan^2 \theta}}{\tan \theta}$	$\sqrt{1 + \cot^2 \theta}$	$\frac{\sec \theta}{\sqrt{\sec^2 \theta - 1}}$	$\operatorname{cosec} \theta$



Value of Trigonometric Ratios in Different Quadrants



For $(90^\circ \pm \theta)$ and $(270^\circ \pm \theta)$, change

- $\sin \theta \Leftrightarrow \cos \theta$,
- $\tan \theta \Leftrightarrow \cot \theta$,
- $\operatorname{cosec} \theta \Leftrightarrow \sec \theta$

For $(180^\circ \pm \theta)$ and $(360^\circ \pm \theta)$

- The trigonometric functions do not change.

A positive or A Negative sign is used according to the quadrant.

$\sin(-x) = -\sin x$	$\cos(-x) = \cos x$
$\tan(-x) = -\tan x$	$\operatorname{cosec}(-x) = -\operatorname{cosec} x$
$\sec(-x) = \sec x$	$\cot(-x) = -\cot x$

Therefore, $\cos$ and $\sec$ are the **Even** trigonometric functions, Whereas $\sin$, $\tan$, cosec , and $\cot$ are the **Odd** trigonometric functions.

Trigonometric Ratios of Some Special Angles

$$(i) \sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$$

$$(ii) \cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ$$

$$(iii) \sin 18^\circ = \frac{\sqrt{5}-1}{4} = \cos 72^\circ$$

$$(iv) \cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4} = \sin 72^\circ$$

$$(v) \sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ$$

$$(vi) \cos 36^\circ = \frac{\sqrt{5}+1}{4} = \sin 54^\circ$$

$$(vii) \sin 22\frac{1}{2}^\circ = \frac{\sqrt{2}-\sqrt{2}}{2}$$

Important Concept to Solve a Specific Type of Question

1. If $A + B = 90^\circ$

(i) $\sin A \cdot \sec B = 1$ or $\sin A = \cos B$	(iv) $\cot A \cdot \cot B = 1$
(ii) $\cos A \cdot \operatorname{cosec} B = 1$ or $\sec A = \operatorname{cosec} B$	(v) $\sin^2 A + \sin^2 B = 1$
(iii) $\tan A \cdot \tan B = 1$ or $\tan A = \cot B$	(vi) $\cos^2 A + \cos^2 B = 1$

2. Formula for Sum and Difference Of Two Angles

(1) $\sin(A+B) = \sin A \cos B + \cos A \sin B$	(5) $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$
(2) $\sin(A-B) = \sin A \cos B - \cos A \sin B$	(6) $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$
(3) $\cos(A+B) = \cos A \cos B - \sin A \sin B$	(7) $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$
(4) $\cos(A-B) = \cos A \cos B + \sin A \sin B$	(8) $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$
$\sin^2 A - \sin^2 B = \sin(A+B) \sin(A-B)$	$\cos^2 A - \cos^2 B = \cos(A+B) \cos(A-B)$

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3. Tangent Formulae:

$\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \tan B)$	$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cot(A + B) = (\cot A \cot B - 1) / (\cot B - \cot A)$	$\cot(A - B) = (\cot A \cot B + 1) / (\cot B - \cot A)$

4. Half, Double, and Triple-Angles Trigonometric Ratios Identities

$\sin 2\theta = 2 \sin\theta \cos\theta$	1. $\cos 2\theta = \cos^2\theta - \sin^2\theta$ 2. $\cos 2\theta = 2 \cos^2\theta - 1$ 3. $\cos 2\theta = 1 - 2 \sin^2\theta$ 4. $\cos 2\theta = (1 - \tan^2\theta) / (1 + \tan^2\theta)$
1. $\tan 2\theta = (2 \tan\theta) / (1 - \tan^2\theta)$ 2. $\cot 2\theta = (\cot\theta - \tan\theta) / 2$	1. $\sec 2\theta = \sec^2\theta / (2 - \sec^2\theta)$ 2. $\operatorname{cosec} 2\theta = (\sec\theta \cdot \operatorname{cosec}\theta) / 2$

5. Function Sum Formulae

- $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right)$
- $\cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right) \sin\left(\frac{D-C}{2}\right)$

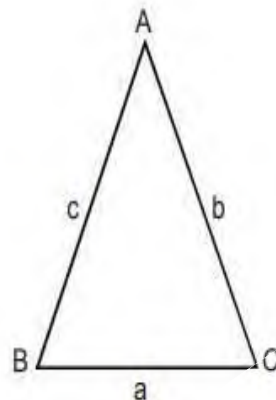
Triple Angle Trigonometric Ratios Identities

- $\sin 3\theta = 3 \sin\theta - 4 \sin^3\theta$
- $\cos 3\theta = 4 \cos^3\theta - 3 \cos\theta$
- $\tan 3\theta = (3 \tan\theta - \tan^3\theta) / (1 - 3 \tan^2\theta)$



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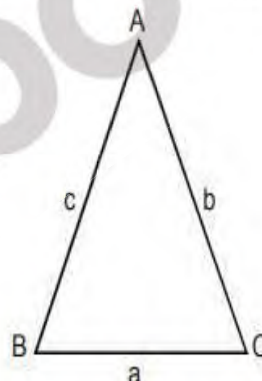
Sine Rule



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Where a, b, and c are the sides of the triangle opposite to the vertex A, B, and C respectively, and, R is the circumradius of the triangle.

Cosine Rule



For a triangle with sides of lengths a, b, and c, where the angle opposite side c is angle C, the Cosine Rule states

$$c^2 = a^2 + b^2 - 2ab \cos(C)$$

Additional Information

- $\sin\theta \cdot \sin(60^\circ - \theta) \cdot \sin(60^\circ + \theta) = 1/4 \sin 3\theta$
- $\cos\theta \cos(60^\circ - \theta) \cdot \cos(60^\circ + \theta) = 1/4 \cos 3\theta$
- $\tan\theta \cdot \tan(60^\circ - \theta) \cdot \tan(60^\circ + \theta) = \tan 3\theta$



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Trigonometric series

1. Sine Series: $\sin(0^\circ) \times \sin(1^\circ) \times \sin(2^\circ) \times \sin(3^\circ) \dots \times \sin(90^\circ) = 0$
2. Cosine Series: $\cos(0^\circ) \times \cos(1^\circ) \times \cos(2^\circ) \times \cos(3^\circ) \times \dots \times \cos(90^\circ) = 0$
3. Tangent Series: $\tan(0^\circ) \times \tan(1^\circ) \times \tan(2^\circ) \times \tan(3^\circ) \times \dots \times \tan(89^\circ) = \infty$
4. Cosecant Series: $\operatorname{cosec}(0^\circ) \times \operatorname{cosec}(1^\circ) \times \operatorname{cosec}(2^\circ) \times \operatorname{cosec}(3^\circ) \times \dots \times \operatorname{cosec}(89^\circ) = \infty$
5. Secant Series: $\sec(0^\circ) \times \sec(1^\circ) \times \sec(2^\circ) \times \sec(3^\circ) \times \dots \times \sec(89^\circ) = \infty$
6. Cotangent Series: $\cot(0^\circ) \times \cot(1^\circ) \times \cot(2^\circ) \times \cot(3^\circ) \times \dots \times \cot(89^\circ) = \infty$

Important Facts Useful for Solving Question on Trigonometry

1. $\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta$
2. $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$
3. If $\cos \theta + \cos^2 \theta + \cos^3 \theta = 1$ then $\sin^6 \theta - 4\sin^4 \theta + 8\sin^2 \theta = 4$
4. If $\sin \theta + \sin^2 \theta + \sin^3 \theta = 1$ then $\cos^6 \theta - 4\cos^4 \theta + 8\cos^2 \theta = 4$
5. $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta) = 2$
6. $(1 + \cot \theta + \sec \theta)(1 + \tan \theta - \operatorname{cosec} \theta) = 2$
7. $(\sin \theta + \cos \theta + 1)(\sin \theta + \cos \theta - 1) = 2\sin \theta \cos \theta$
8. $\frac{\sec \theta + \tan \theta - 1}{\tan \theta - \sec \theta + 1} = \sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 - \sin \theta}$
9. $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{1}{\sec \theta - \tan \theta} = \sec \theta + \tan \theta$
10. $\frac{1 + \sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} = 2\sec \theta$
11. $\frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$
12. $\frac{\cot \theta}{\operatorname{cosec} \theta - 1} = \frac{\operatorname{cosec} \theta - 1}{\cot \theta}$
13. $\sec \theta - \tan \theta = k$ i.e., $\sec \theta + \tan \theta = \frac{1}{k}$
 $\Rightarrow \operatorname{cosec} \theta + \cot \theta = \frac{1+k}{1-k}$
14. $(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}, 0^\circ < \theta < 90^\circ$
15. $\frac{\tan \theta_1 + \cot \theta_2}{\tan(90^\circ - \theta_1) + \cot(90^\circ - \theta_2)} = \tan \theta_1 \cdot \cot \theta_2$
16. $\tan \theta (1 + \sec 2\theta)(1 + \sec 4\theta)(1 + \sec 8\theta) = \tan 8\theta$
17. $\sec^4 \theta - \tan^4 \theta = \sec^2 \theta + \tan^2 \theta = 1 + 2\tan^2 \theta = 2\sec^2 \theta - 1$
18. $\operatorname{cosec}^4 \theta - \cot^4 \theta = \operatorname{cosec}^2 \theta + \cot^2 \theta = 1 + 2\cot^2 \theta = 2\operatorname{cosec}^2 \theta - 1$
19. $\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} = \frac{1 + \tan \theta}{1 - \tan \theta} = \tan\left(\frac{\pi}{4} + \theta\right)$



$$20. a \cos \theta - b \sin \theta = x,$$

$$a \sin \theta + b \cos \theta = c$$

$$\Rightarrow x = \pm \sqrt{a^2 + b^2 - c^2}$$

$$21. a \sec \theta + b \tan \theta = c \quad a \tan \theta + b \sec \theta = x$$

$$\Rightarrow x = \pm \sqrt{c^2 + b^2 - a^2}$$

$$22. a \sec \theta - b \tan \theta = c \quad a \tan \theta - b \sec \theta = x$$

$$\Rightarrow x = \pm \sqrt{c^2 + b^2 - a^2}$$

$$23. \sec \theta + \tan \theta = \frac{a}{b} \quad \operatorname{cosec} \theta + \cot \theta = \frac{a+b}{a-b}$$

24. (i).

(a)

$$\sin 1^\circ \cdot \sin 2^\circ \cdot \sin 3^\circ \cdot \sin 4^\circ \dots \sin 180^\circ = 0$$

$$(b) \sin 1^\circ \cdot \sin 2^\circ \cdot \sin 3^\circ \cdot \sin 4^\circ \dots \sin 180^\circ = 0$$

(ii).

$$(a) \cos 1^\circ \cdot \cos 2^\circ \dots \cos 90^\circ = 0$$

$$(b) \cos 1^\circ \cdot \cos 2^\circ \dots \cos 90^\circ = 0$$

(iii)

$$(a) \tan 1^\circ \cdot \tan 2^\circ \dots \tan 89^\circ = 1$$

$$(iv) \sin^2 \theta + \sin^2 (\theta + x) + \dots$$

$$+ \sin^2 [90 - (\theta + x)] + \sin^2 (90 - \theta) = \frac{\text{Total terms}}{2}$$

$$(v) \text{ If } \sec \theta \pm \tan \theta = x, \text{ then } \sec \theta = \frac{x^2 + 1}{2x}$$

$$(vi) \text{ If } \sin \theta + \cos \theta = x, \text{ then}$$

$$\sin \theta - \cos \theta = \sqrt{2 - x^2}$$

$$(vii) \text{ If } \sin \theta + \operatorname{cosec} \theta = x, \text{ then}$$

$$\sin^n \theta + \operatorname{cosec}^n \theta = x$$

$$(viii) \text{ If } \tan \theta + \cot \theta = x, \text{ then}$$

$$\tan^n \theta + \cot^n \theta = x$$

(xi) It should be noted that

$$\sin^2 \theta = (\sin \theta)^2, \sin^3 \theta = (\sin \theta)^3, \cos^3 \theta = (\cos \theta)^3, \text{ etc.}$$

Measure of an angle

$$180^\circ = \pi \text{ radians}$$

$$\text{Therefore, } 1^\circ = \frac{\pi}{180^\circ} \text{ radian}$$

Ex:

$$30^\circ = \frac{\pi}{180^\circ} \times 30^\circ = \frac{\pi}{6}$$

Degree	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

Important Points

1. The angle between two consecutive digits in a clock is radian.
2. The hour hand rotates through an angle of 30° in one hour i.e. in one minute.
3. The minute hand rotates through an angle of 6° in one minute.



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Maxima & Minima

Trigonometric Function	Maximum Values	Minimum Values
Sine	1	-1
Cosine	1	-1
Tangent	∞	$-\infty$
Cotangent	∞	$-\infty$
Secant	∞	$-\infty$
Cosecant	∞	$-\infty$

1. $a \sin^2 \theta + b \cos^2 \theta$

- If $a > b$ then, $\max = a$ and $\min = b$
- If $a < b$ then, $\max = b$ and $\min = a$.

2. $\sin^n \theta \cos^n \theta$

- If $n = \text{odd}$ then, $\max = \frac{1}{2n}$
and $\min = -\frac{1}{2n}$
- If $n = \text{even}$ then, $\max = \frac{1}{2n}$
and $\min = 0$.

3. $a \sin \theta + b \cos \theta \rightarrow \max =$

$+\sqrt{a^2 + b^2}$ and $\min =$

$-\sqrt{a^2 + b^2}$

4. $a \sin^2 \theta + b \cos^2 \theta \rightarrow \min = 2\sqrt{ab}$
(when $a > b$)

$a \cos^2 \theta + b \sec^2 \theta \rightarrow \min$
 $= a + b$ (when $a < b$) and
 $\max = \infty$

5. $a \sec^2 \theta + b \csc^2 \theta \rightarrow \min$
 $(\sqrt{a} + \sqrt{b})^2$

6. $\sin^{2m} \theta + b \cos^{2n} \theta \rightarrow \max = +1$
 m, n are natural numbers.

7. If $y = \cos^2 \theta + \sec \theta$, then $y \geq 2$.

8. $\sin^{2n} \theta + \cos^{2n} \theta \rightarrow \max = 1$, and
 $\min \text{ value} = \text{put } \theta \rightarrow 45^\circ$

9. $a \tan^2 \theta + b \cot^2 \theta \rightarrow \min = 2\sqrt{ab}$
and $\max = \infty$.



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Heights & Distances

In trigonometry, height is the opposite side in a right triangle, and distance is the hypotenuse. They're related through trigonometric ratios.

Object:

A thing or a person seen by someone can be said to be an object with respect to the viewer.

Viewer:

The person who sees the object.

Line of Sight:

An imaginary line joining the object and the viewer's eyes is called the line of sight.

Horizontal Line:

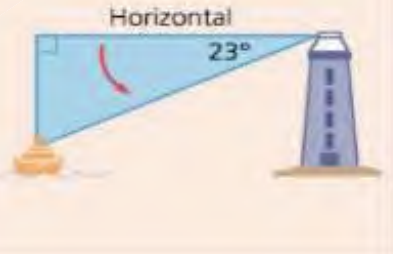
All the lines parallel to the surface of the earth are called horizontal lines.

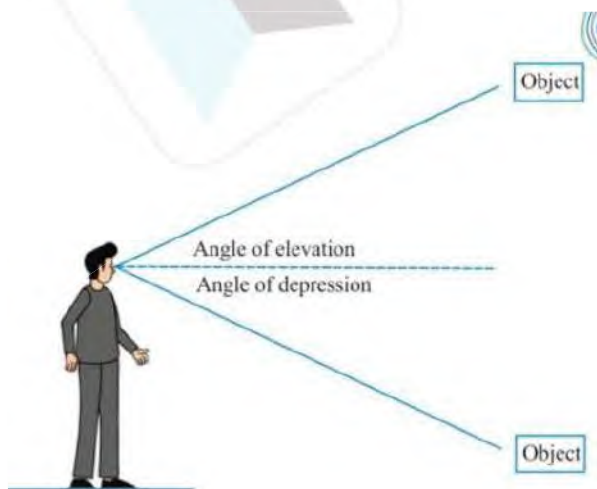
Angle of Inclination:

When an object is present above the eye level of the viewer, then the angle between the horizontal (eye level) and the line of sight is called the angle of elevation.

Angle of Depression:

When an object is present below the eye level of the viewer, then the angle between the horizontal (eye level) and the line of sight is called the angle of depression.

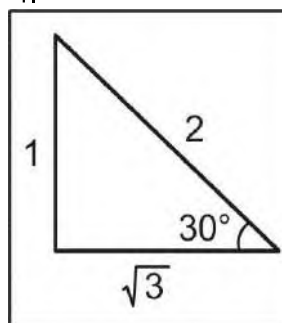
Angle of inclination	Angle of elevation	Angle of depression
Formed between the vertical or horizontal as defined by the situation. The angle of inclination with the vertical is 42° .	Formed by looking up from the horizontal The angle of elevation is 36° .	Formed by looking down from the horizontal. The angle of depression is 23° .
		



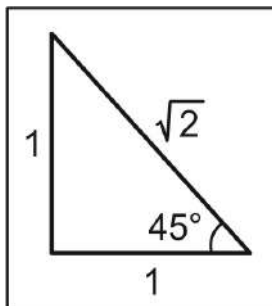
Some Basic Triangles:

For one viewer - one object

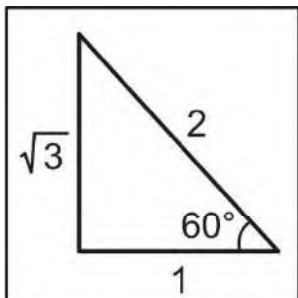
1.



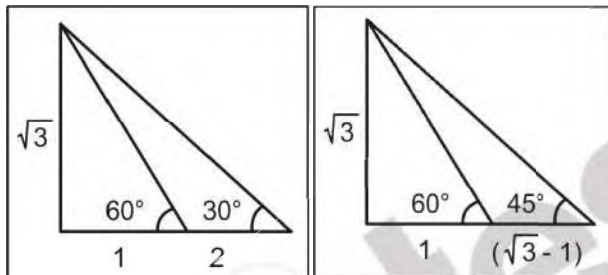
2.



3.

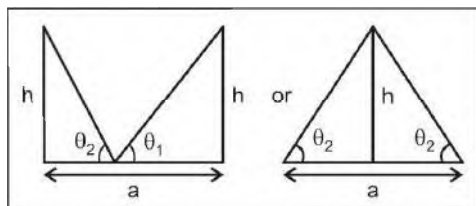
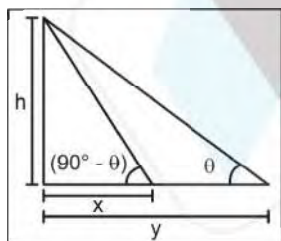


4. or one viewer - two object



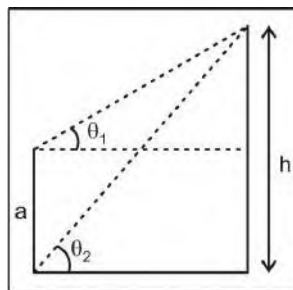
5. If both angles of elevation are complementary

$$h^2 = xy \text{ or } h = \sqrt{xy}$$



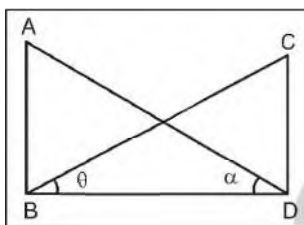
$$a = h(\cot\theta_2 + \cot\theta_1)$$

5.



$$h = \frac{a \cot\theta_1}{\cot\theta_1 - \cot\theta_2}$$

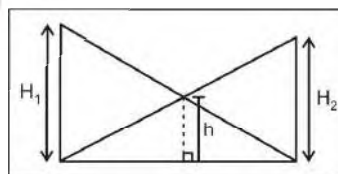
6.



(i) $\theta + \alpha = 90^\circ$ or both the angle of elevation are complementary, then

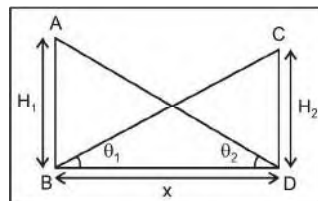
$$x^2 = H_1 H_2 \text{ or } x = \sqrt{H_1 H_2}$$

(ii) Height of the point of intersection of the lines joining from foot of one to top of the other from horizontal line is h.

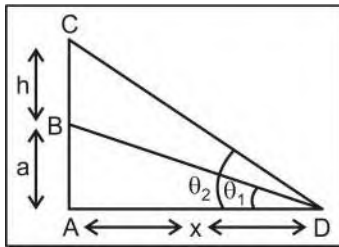


$$\frac{1}{h} = \frac{1}{H_1} + \frac{1}{H_2}$$

(iii) If both angles of elevation are not complementary, then we will solve these problem by using the right-angled triangles ABC and BCD



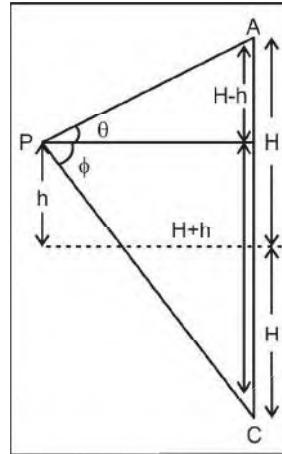
1. First Methods : In this method we use the two right-angled triangles separately.
2. Second Method : Given $\angle ADB = \theta_1$, and $\angle ADC = \theta_2$



If h will be given, x is asked, then

$$\frac{a}{h} = \frac{\tan \theta_1}{\tan \theta_2 - \tan \theta_1} \text{ and } h = x (\tan \theta_2 - \tan \theta_1)$$

The angle of elevation of a cloud at height h above the level of water in a lake is θ and the angle of the depression of its image in the lake is ϕ . Then the height of the cloud above the surface of the lake is : s



$$H = h \left(\frac{\cot \theta + \cot \phi}{\cot \theta - \cot \phi} \right)$$



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